APPENDIX A - 1

INTRODUCTION

AND

PROBLEM DETERMINATION

(GEK - 92762A)

APPENDIX A - 1

TROUBLESHOOTING MANUAL FOR PROPULSION EQUIPMENT

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A-1 INTRODUCTION & PROBLEM DETERMINATION

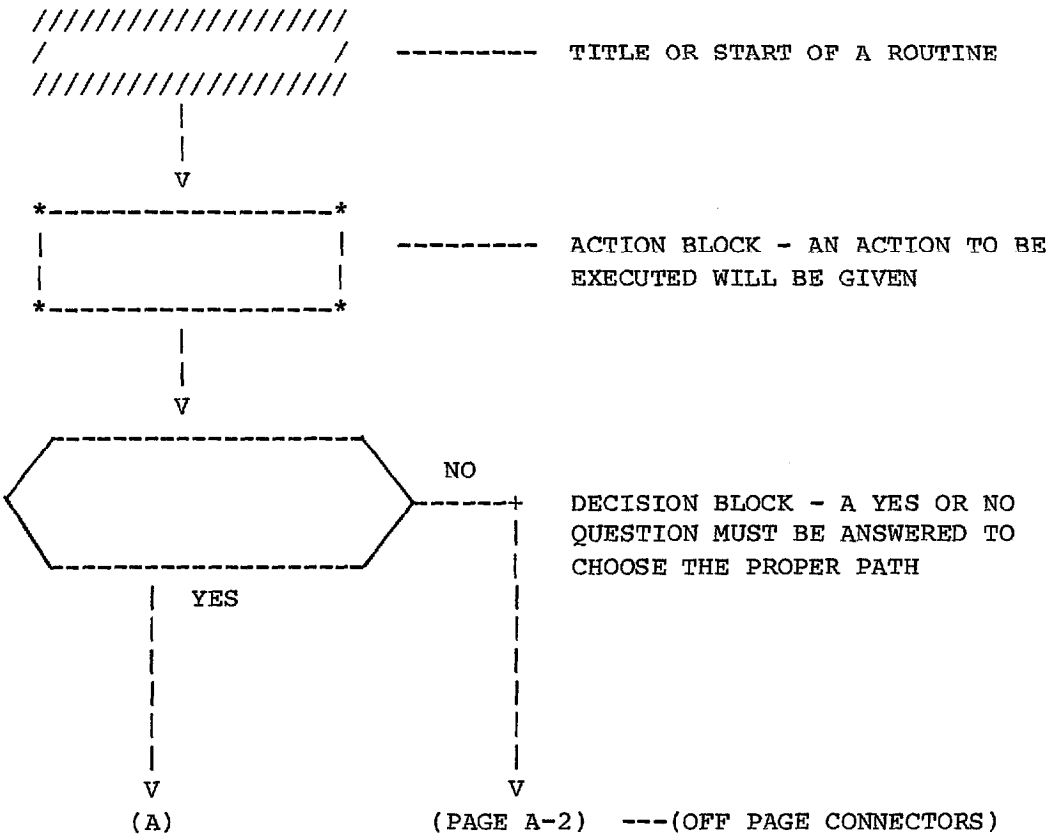
A-1.1 Introduction

The purpose of this document is to aid the electrical maintenance personnel in narrowing down a fault or problem to a particular area. It will cover the majority of possible problems which may occur.

It cannot be a substitute for familiarity with the equipment and/or trouble - shooting tools.

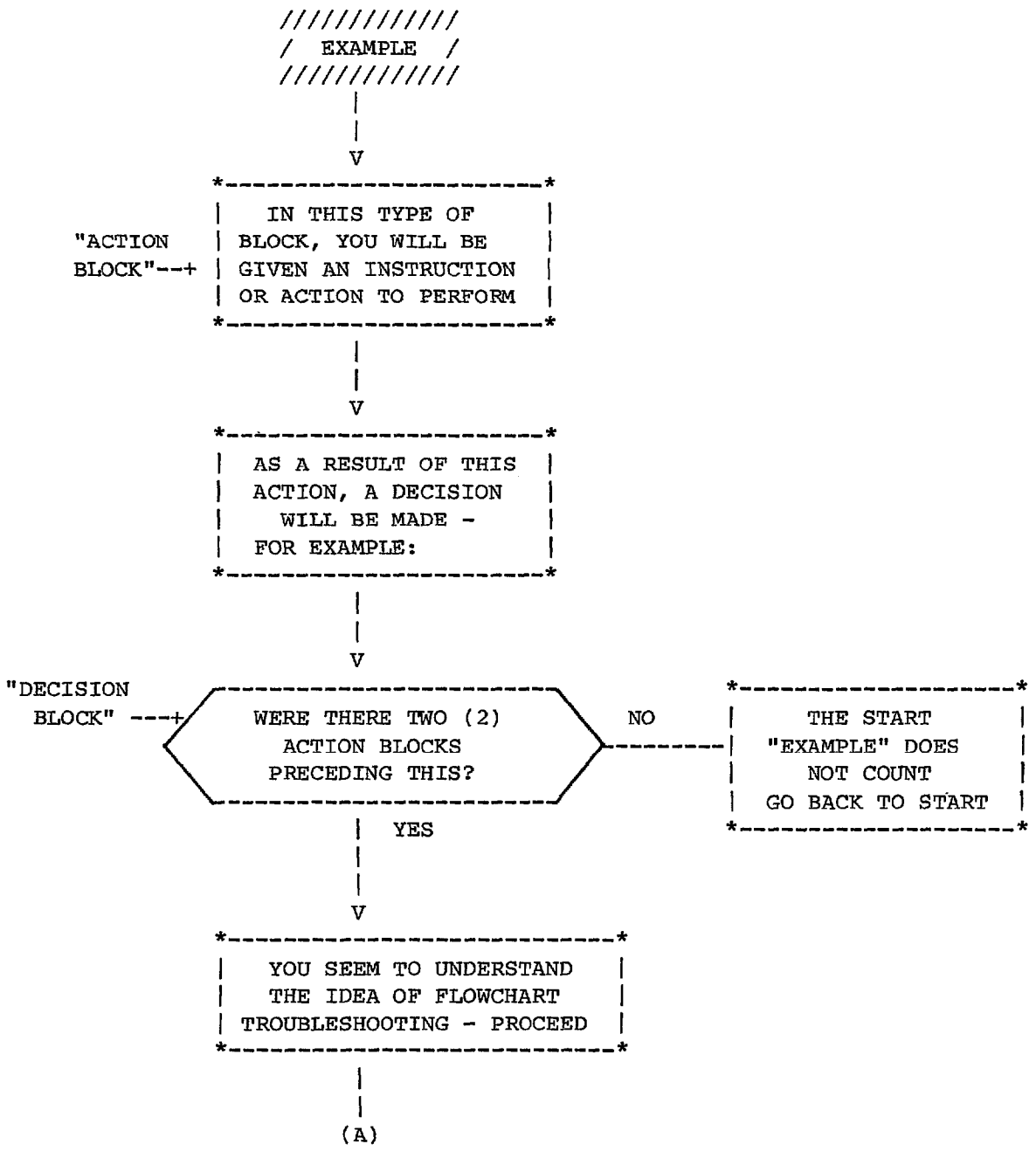
Like any tool, proper use and understanding of that tool will make the user more efficient at the task being performed. Therefore, for a complete understanding of this document, the entire introduction should be read.

Troubleshooting will be performed using a flowchart approach. Given below are the symbols used throughout this document. Following that, an example is given to aid in the understanding of the flowchart method for those who have never used it. (A "+" symbol is used to indicate flow direction change.)

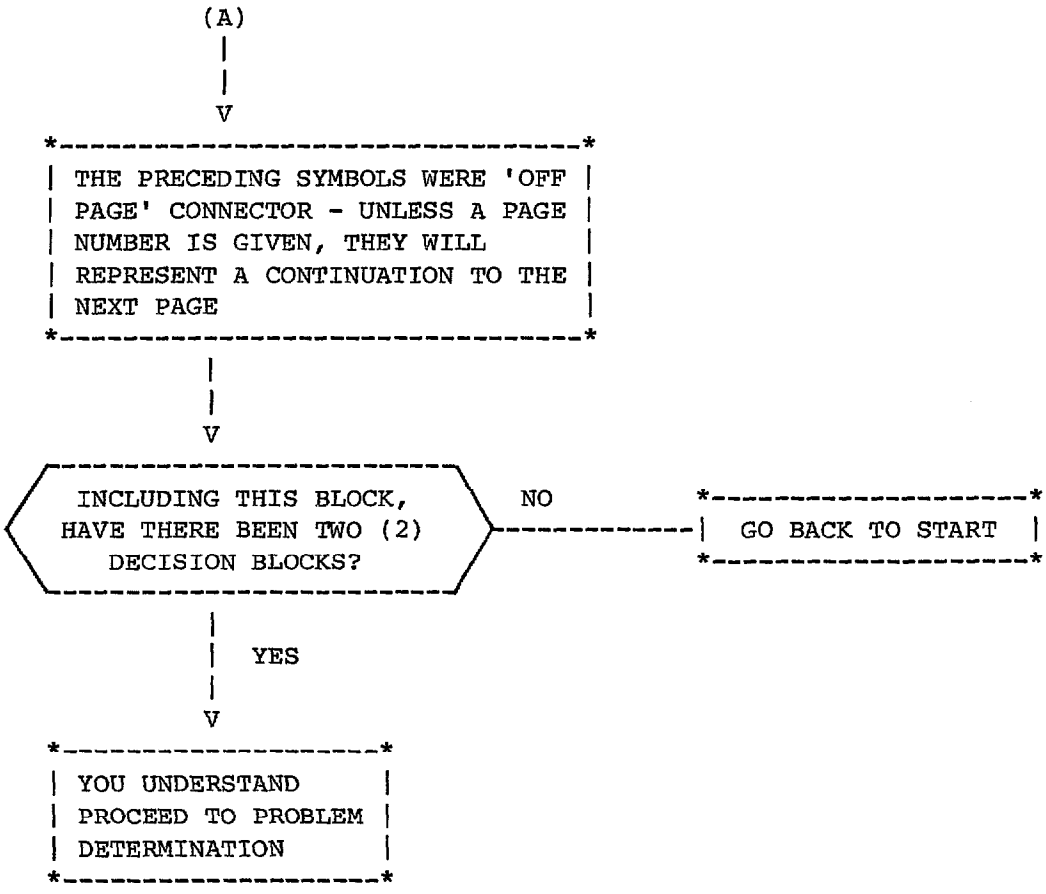


A-1.1 Introduction (Continued)

The following flowchart will give you an introduction into flowchart trouble - shooting. This type of flowchart will be used throughout this manual so understanding it will be essential.



A-1.1 Introduction (Continued)

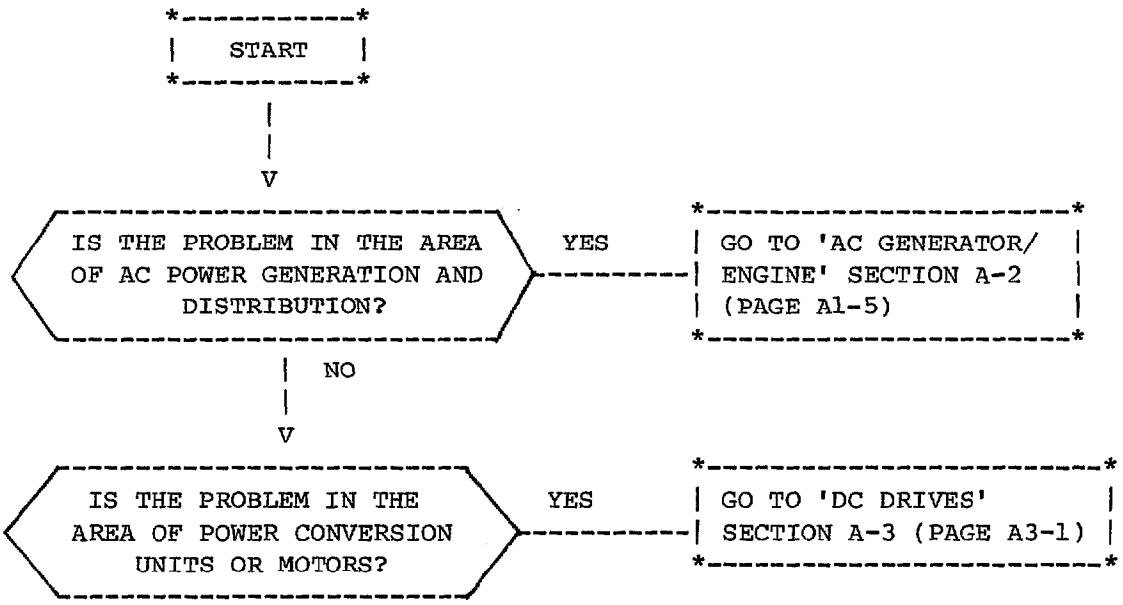


A-1.2 Problem Determination

The electrical system can be broken down into two (2) basic categories.

- 480 VAC Power Generation and Distribution (Generator/Engine Control)
- DC Power (SCR Power Converters and Motors)

In the same sense, problems can be categorized into the same areas. Each category has multiple areas under each topic and the areas sometimes overlap (such as the interaction of the DC and AC control modules to limit loading of the generator/engines). These areas will be taken into account in the course of the flowchart troubleshooting guide. Using the following flowchart, narrow the problem down to a specific area.



APPENDIX A - 2

AC GENERATOR/ENGINE

(GEK - 92762A - Part 2)

APPENDIX A - 2

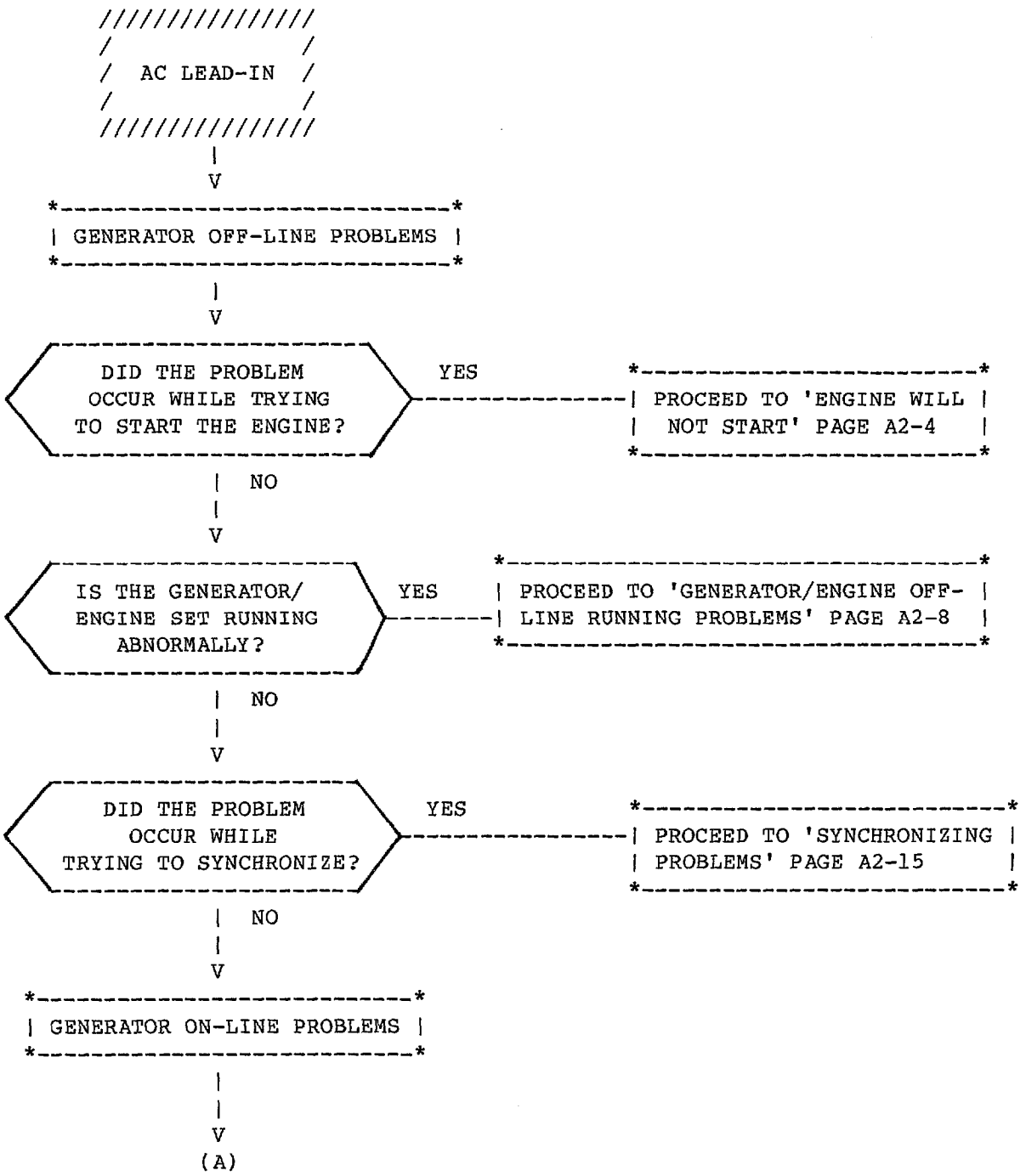
TROUBLESHOOTING MANUAL FOR PROPULSION EQUIPMENT

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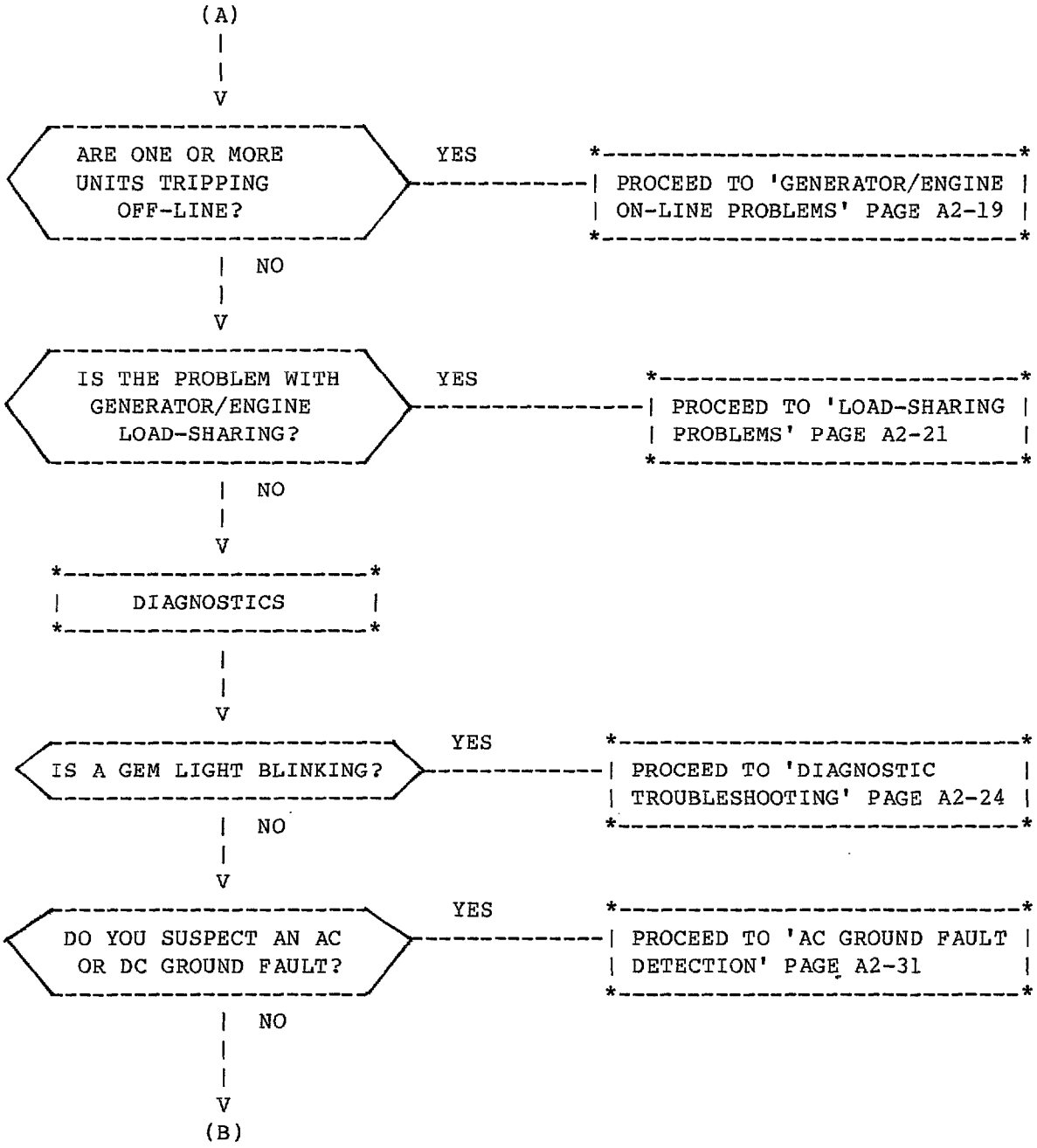
SECTION	TITLE	PAGE
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A-2 AC GENERATOR/ENGINE

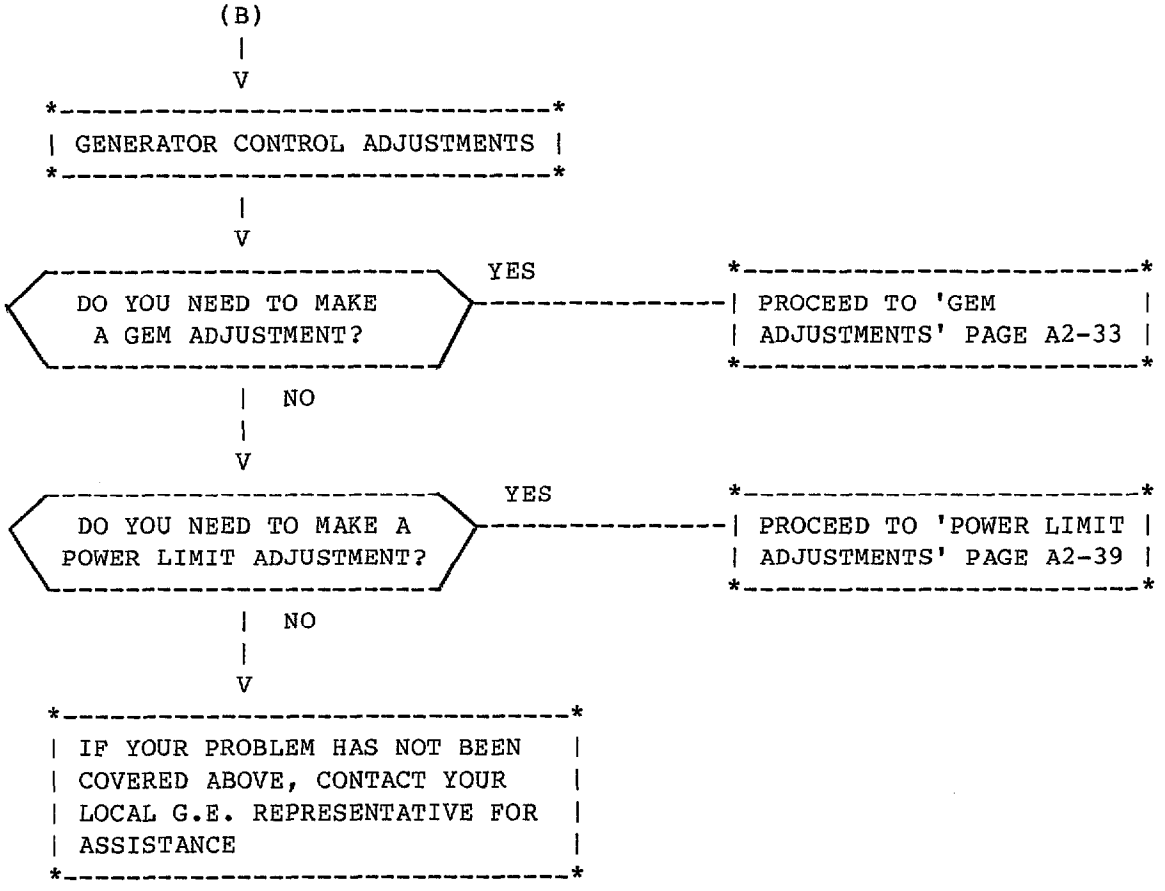
A-2.1 AC Lead-In



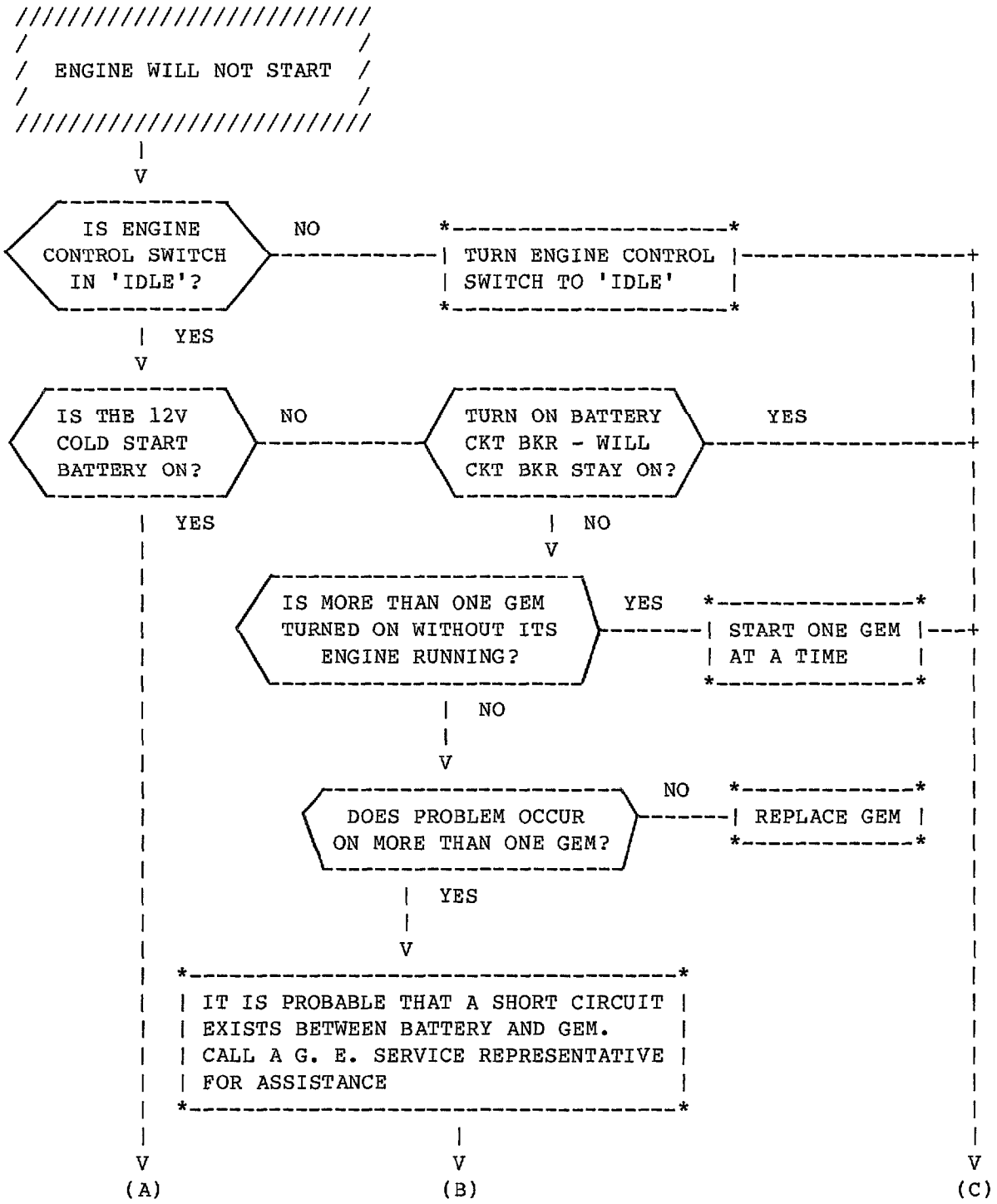
A-2.1 AC Lead-In (Continued)



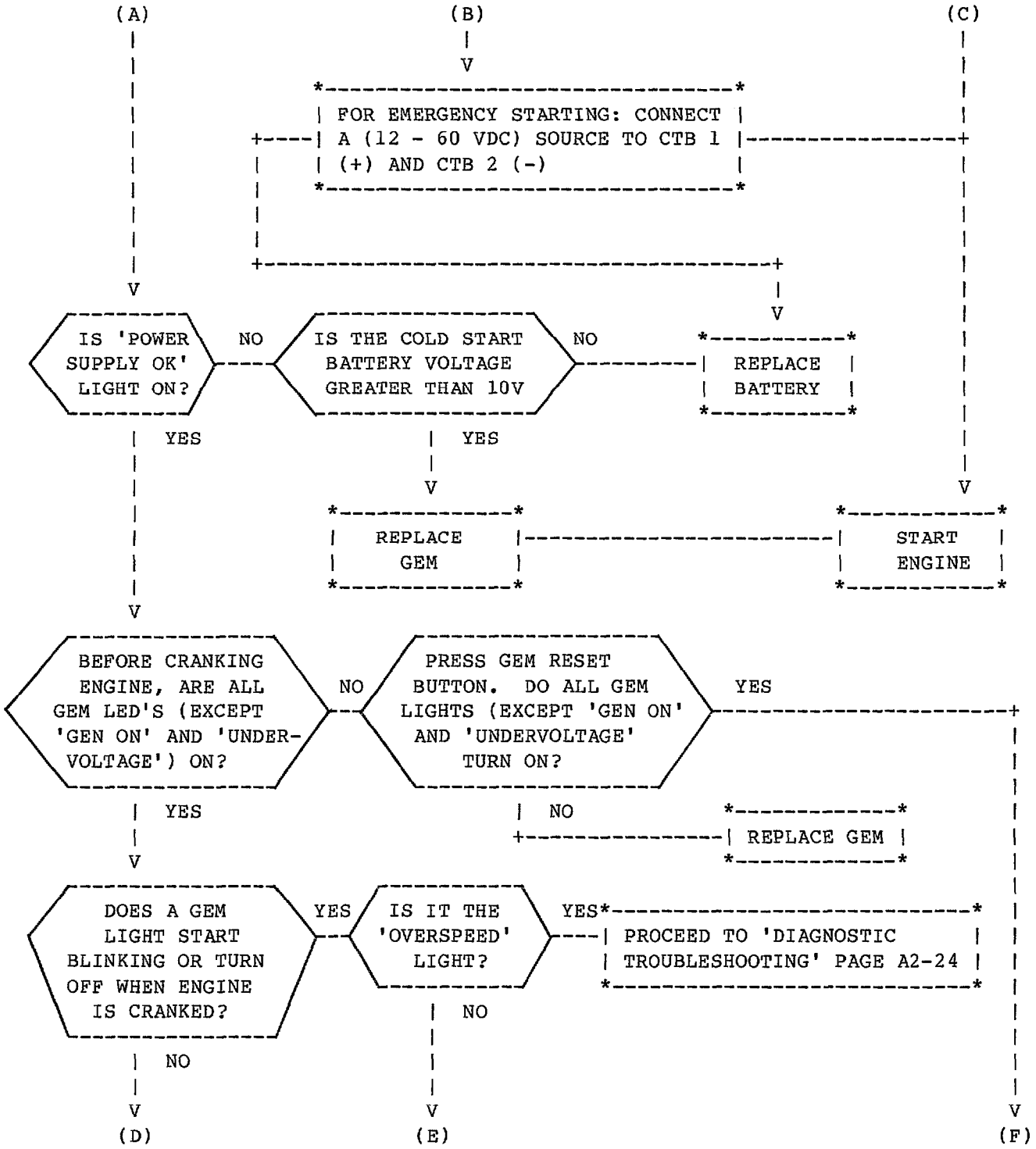
A-2.1 AC Lead-In (Continued)



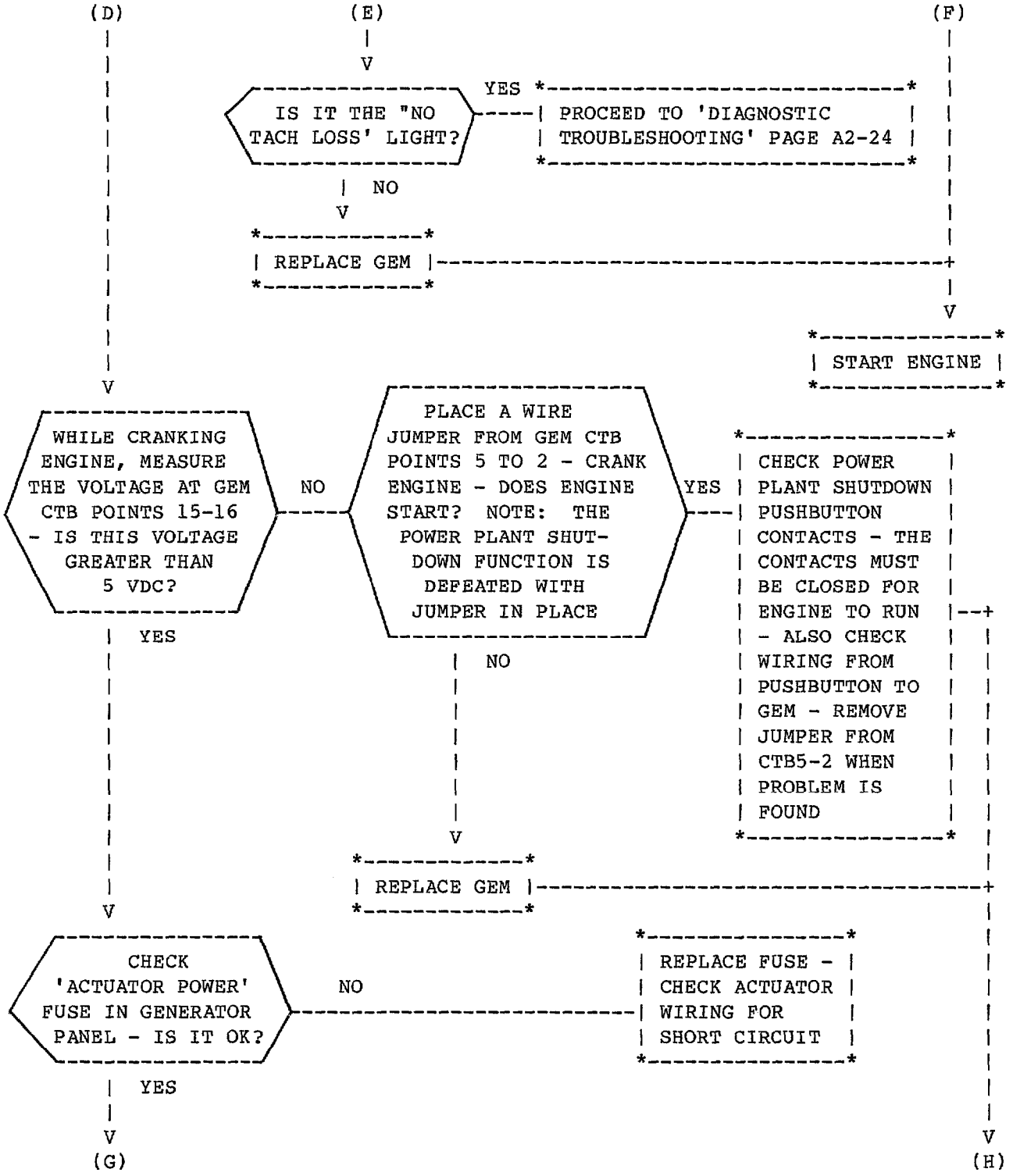
A-2.2 Engine Starting



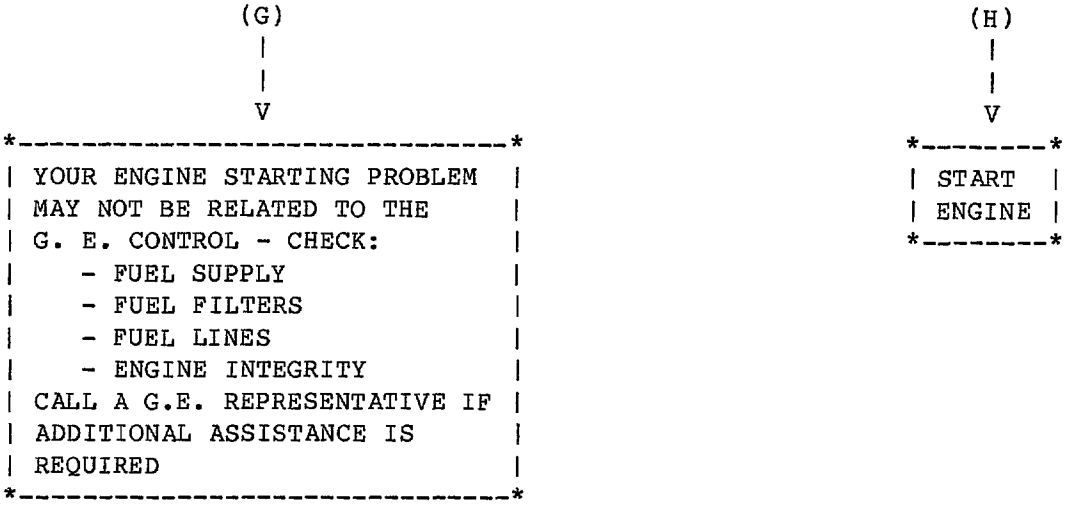
A-2.2 Engine Starting (Continued)



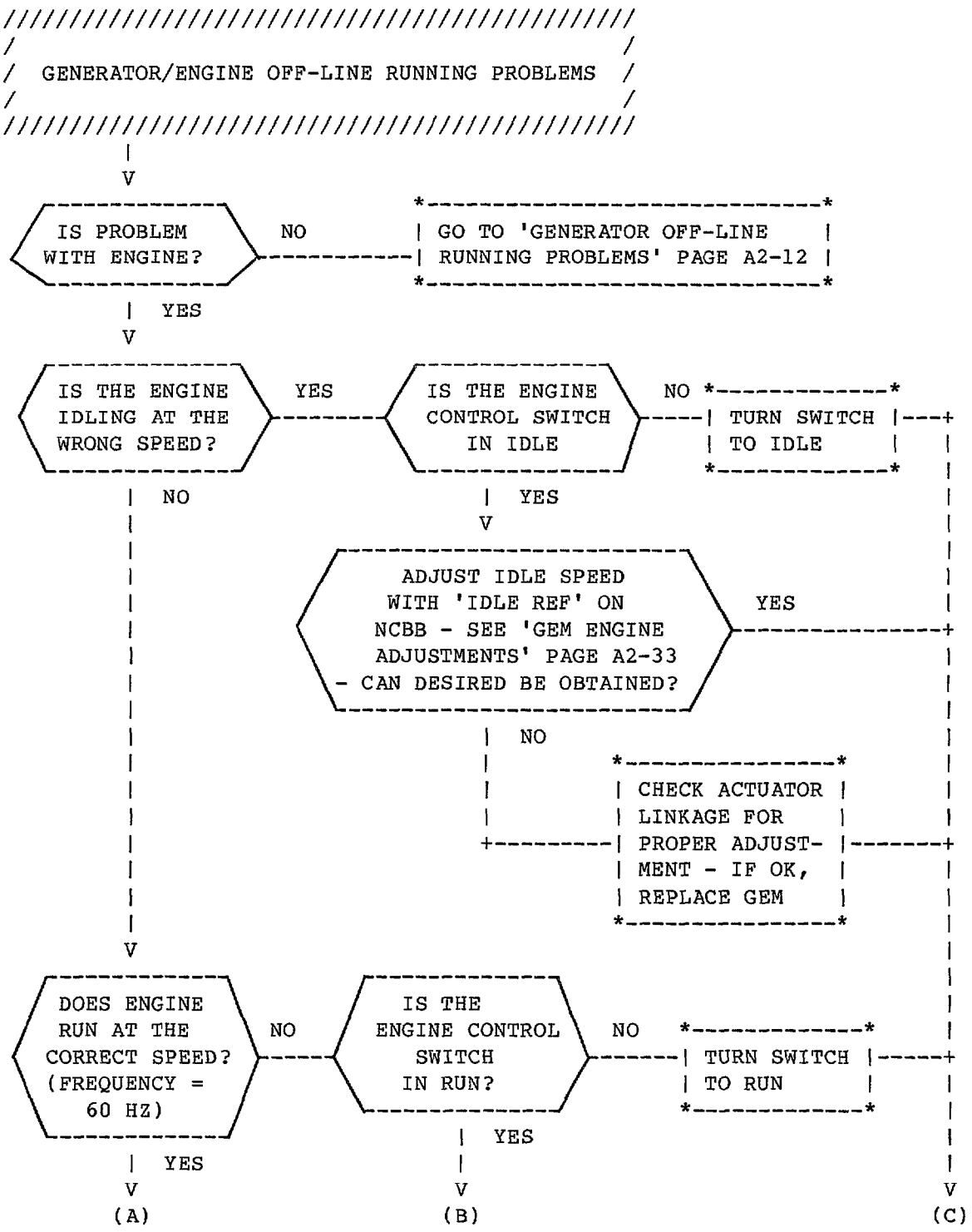
A-2.2 Engine Starting (Continued)



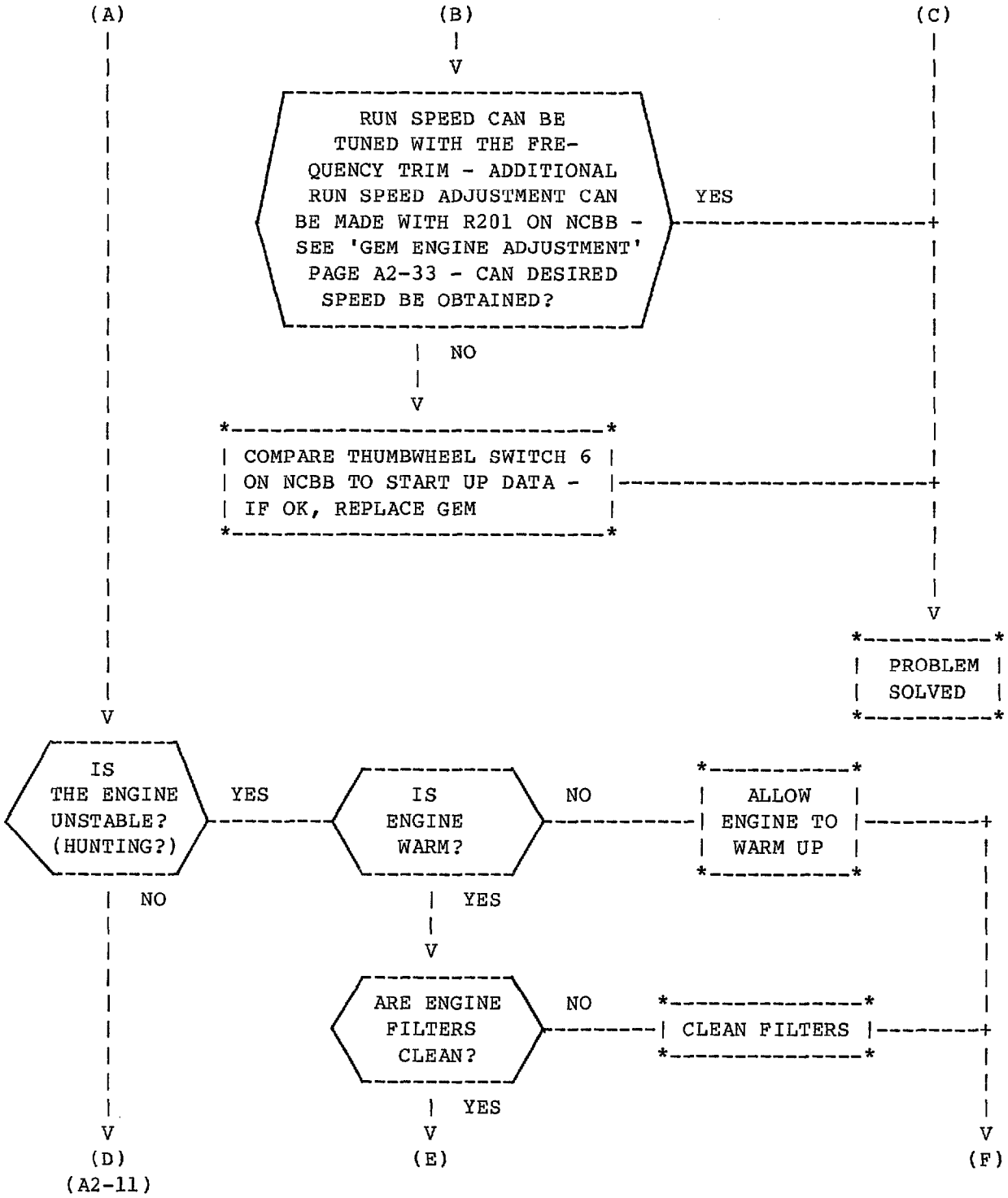
A-2.2 Engine Starting (Continued)



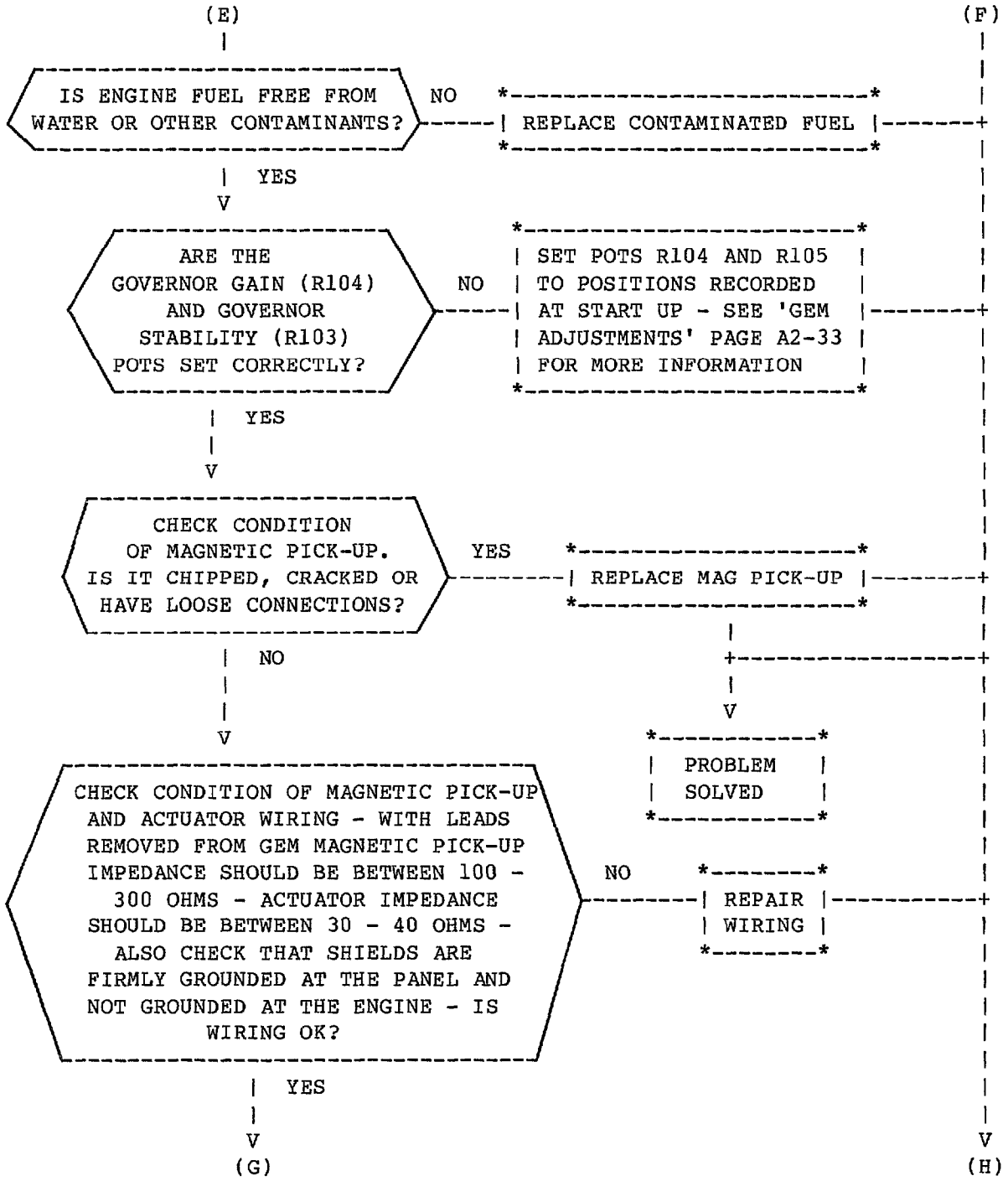
A-2.3 Generator/Engine Off-Line Running Problems



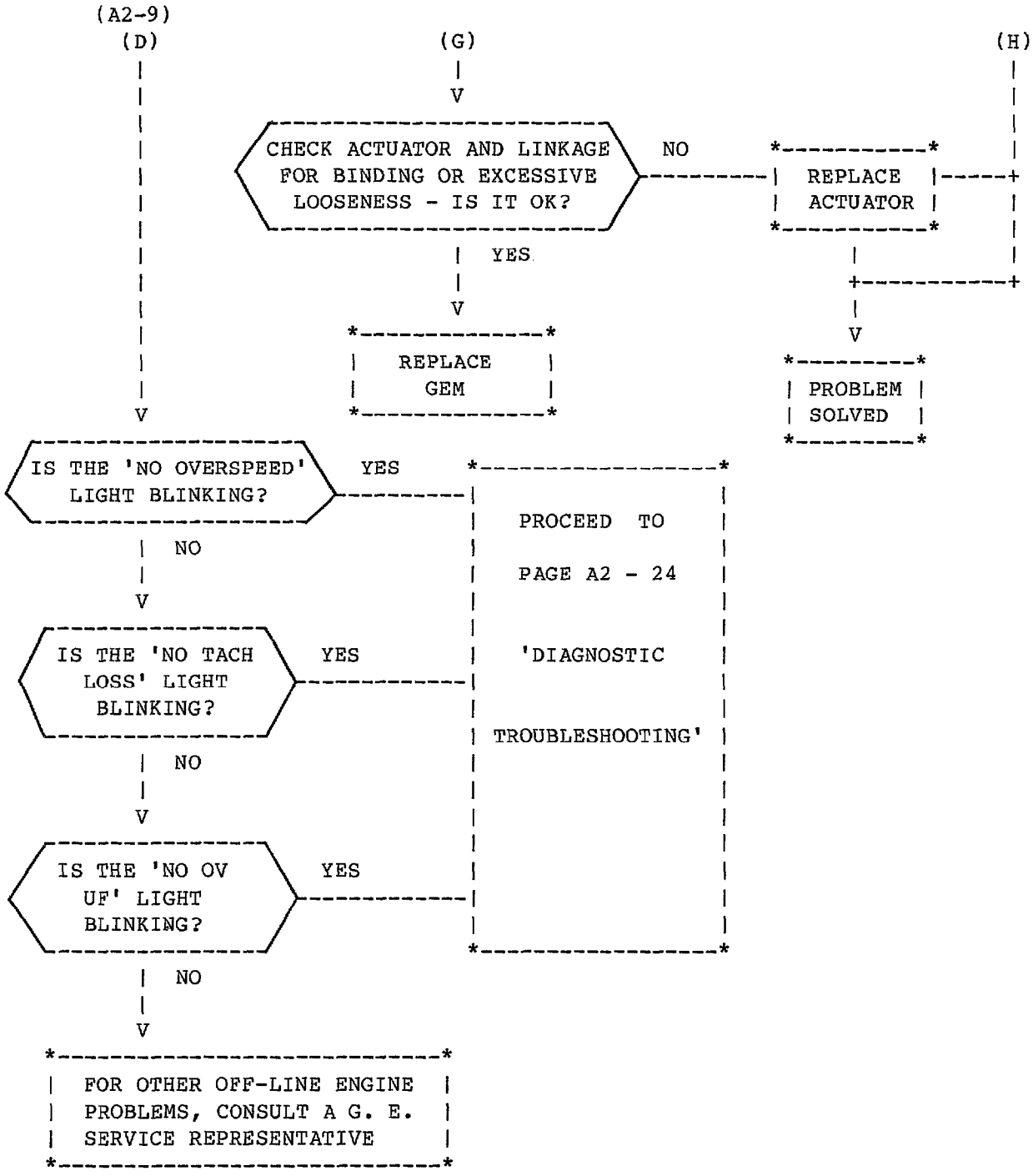
A-2.3 Generator/Engine Off-Line Running Problems (Continued)



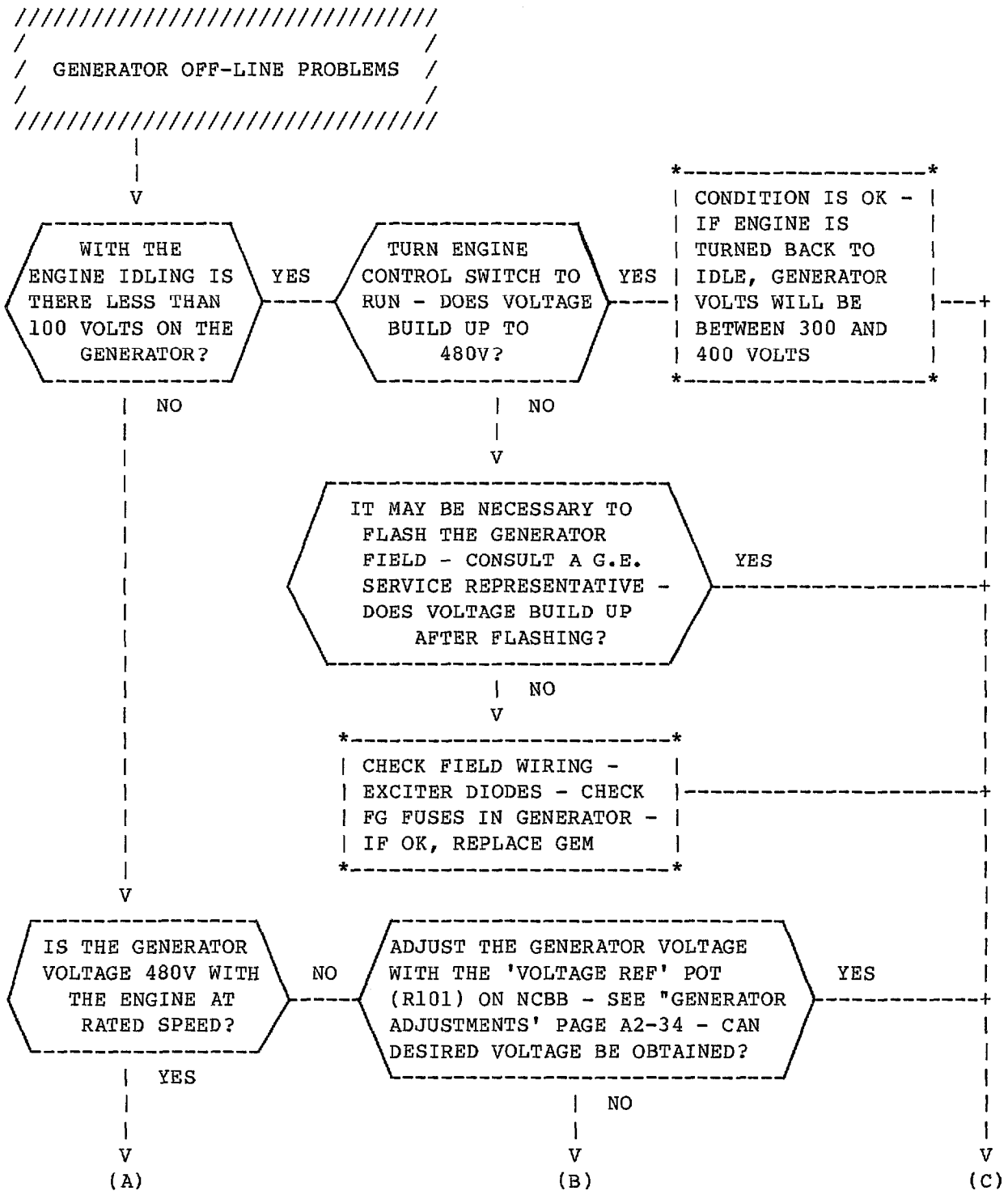
A-2.3 Generator/Engine Off-Line Running Problems (Continued)



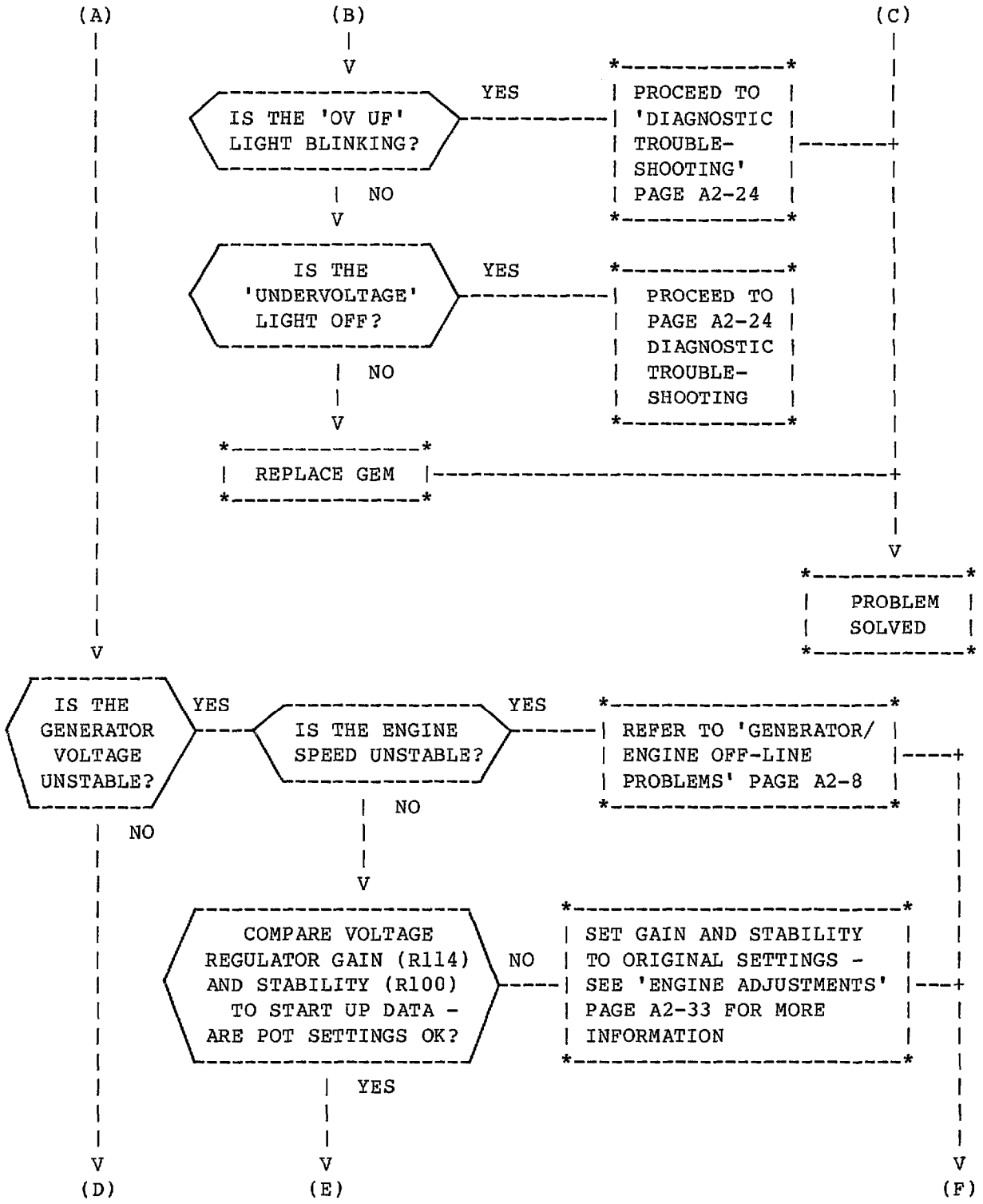
A-2.3 Generator/Engine Off-Line Running Problems (Continued)



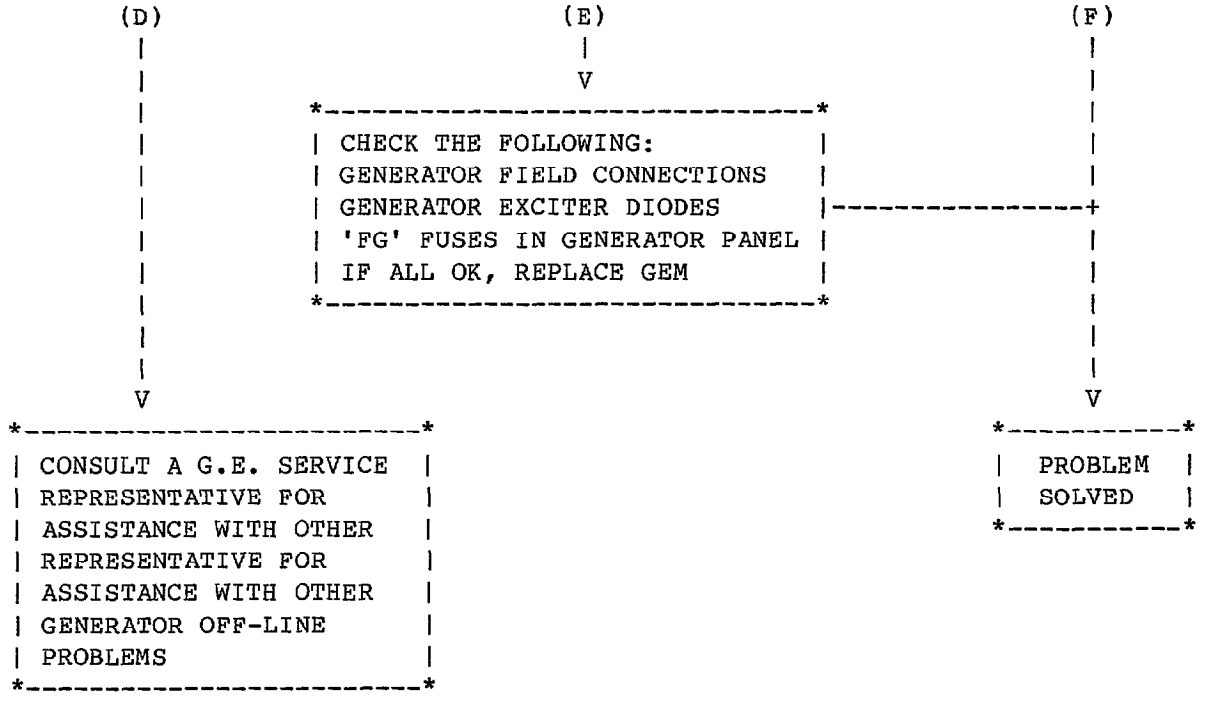
A-2.4 Generator Off-Line Problems



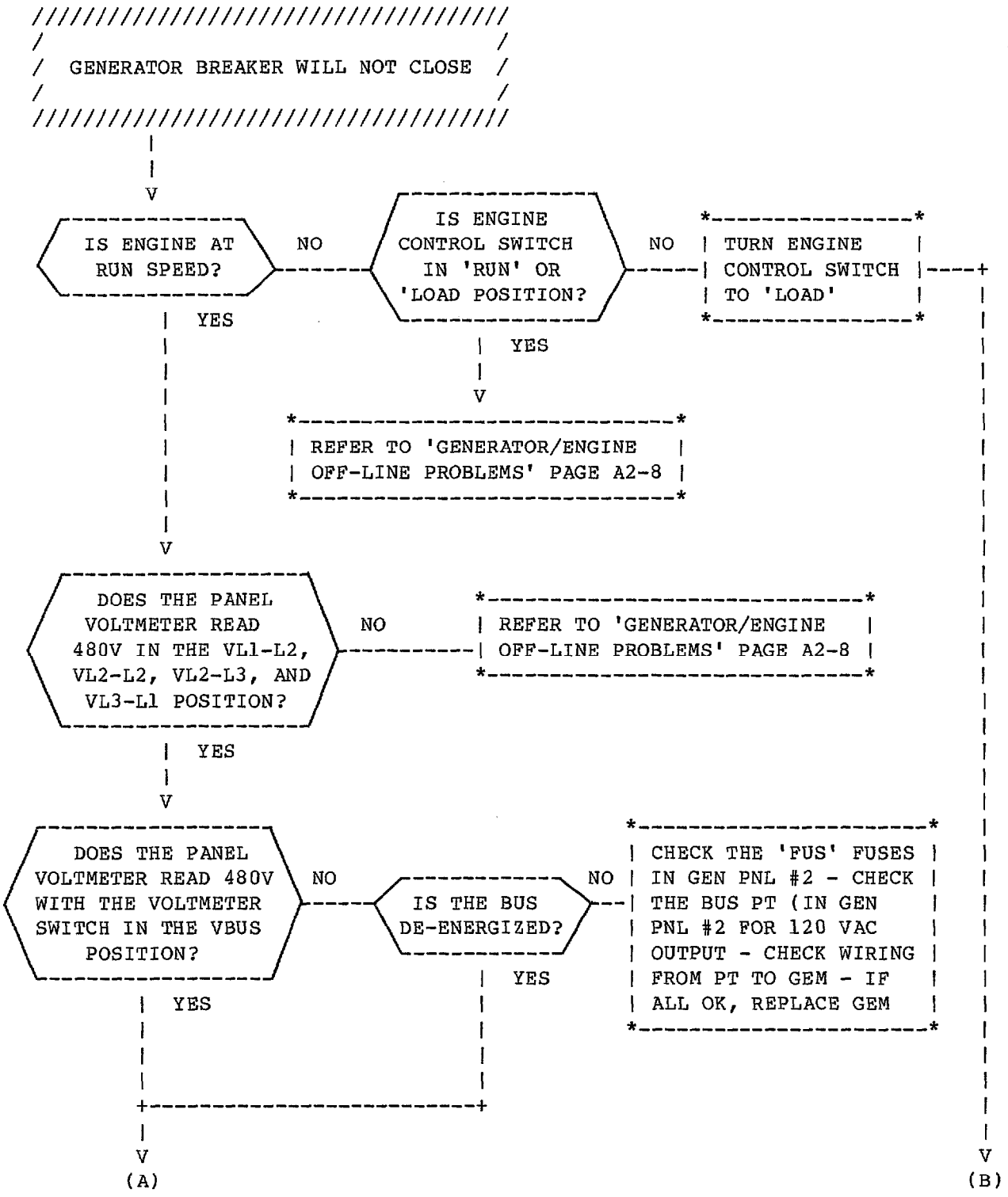
A-2.4 Generator Off-Line Problems (Continued)



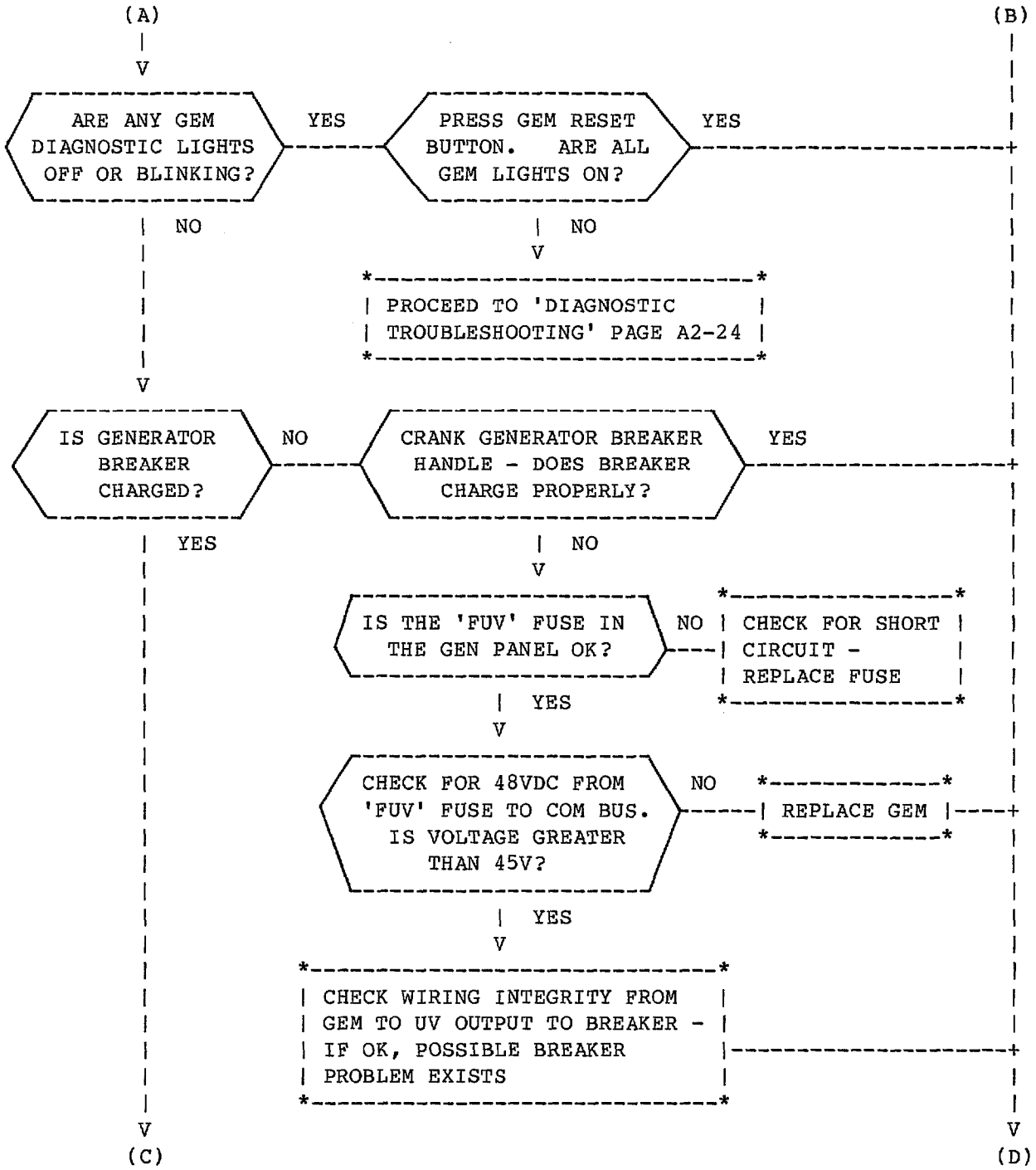
A-2.4 Generator Off-Line Problems (Continued)



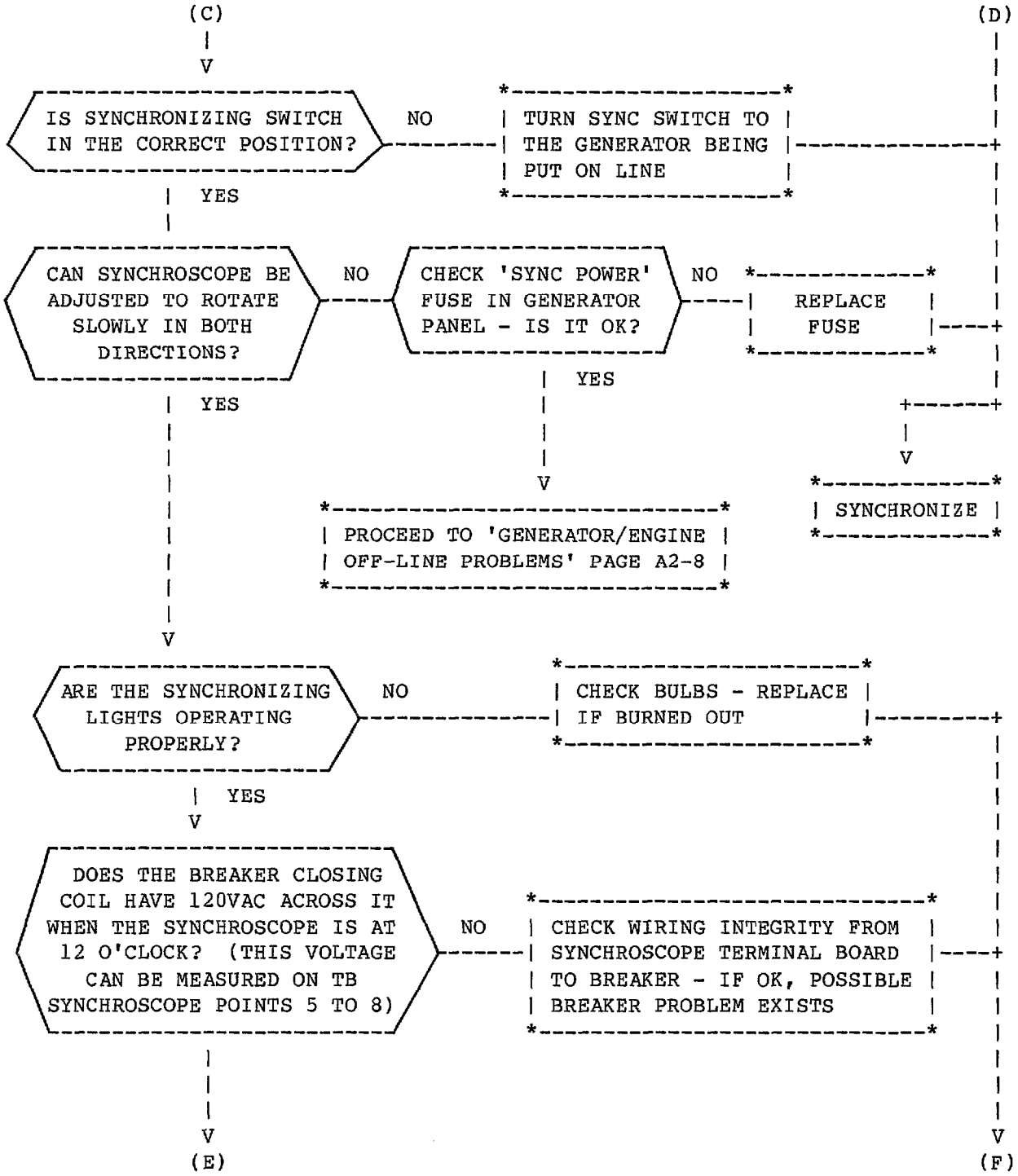
A-2.5 Synchronizing Problems



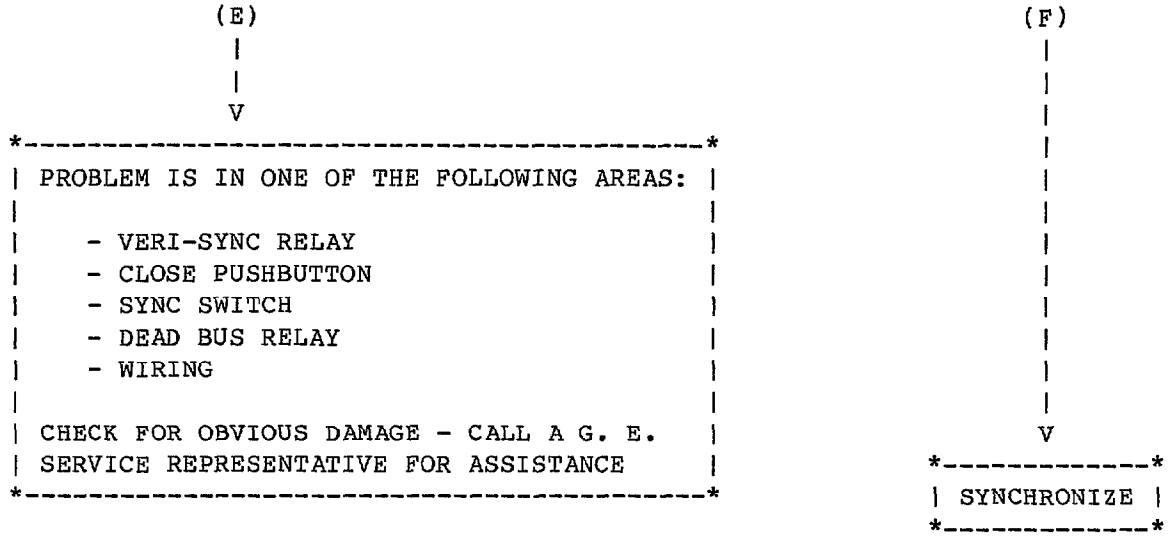
A-2.5 Synchronizing Problems (Continued)



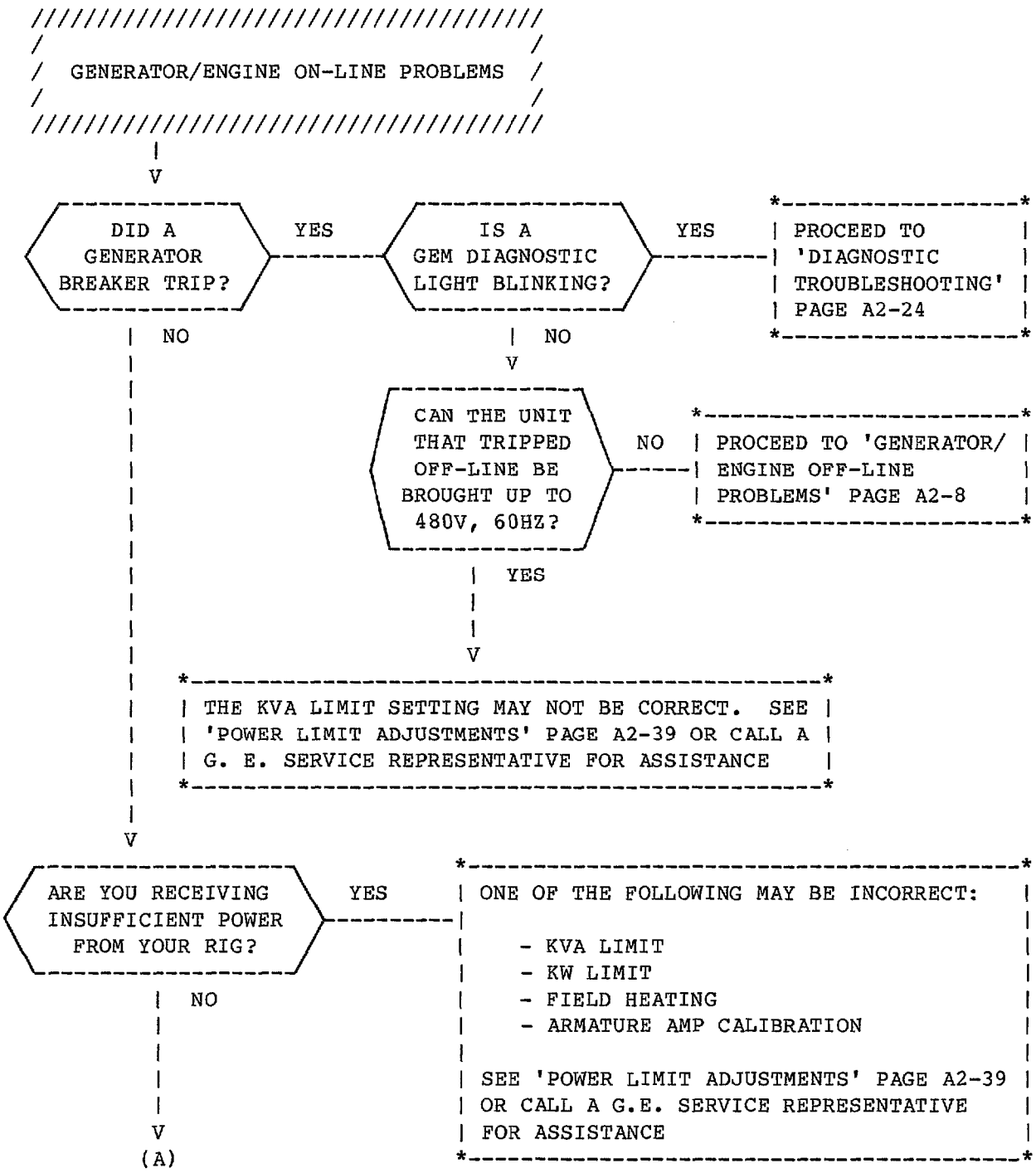
A-2.5 Synchronizing Problems (Continued)



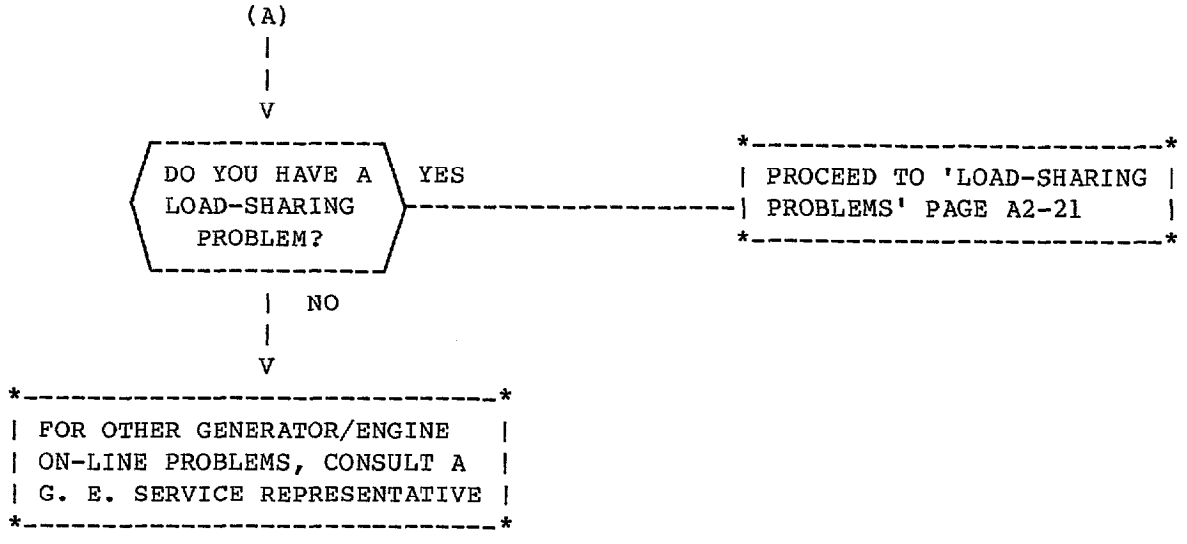
A-2.5 Synchronizing Problems (Continued)



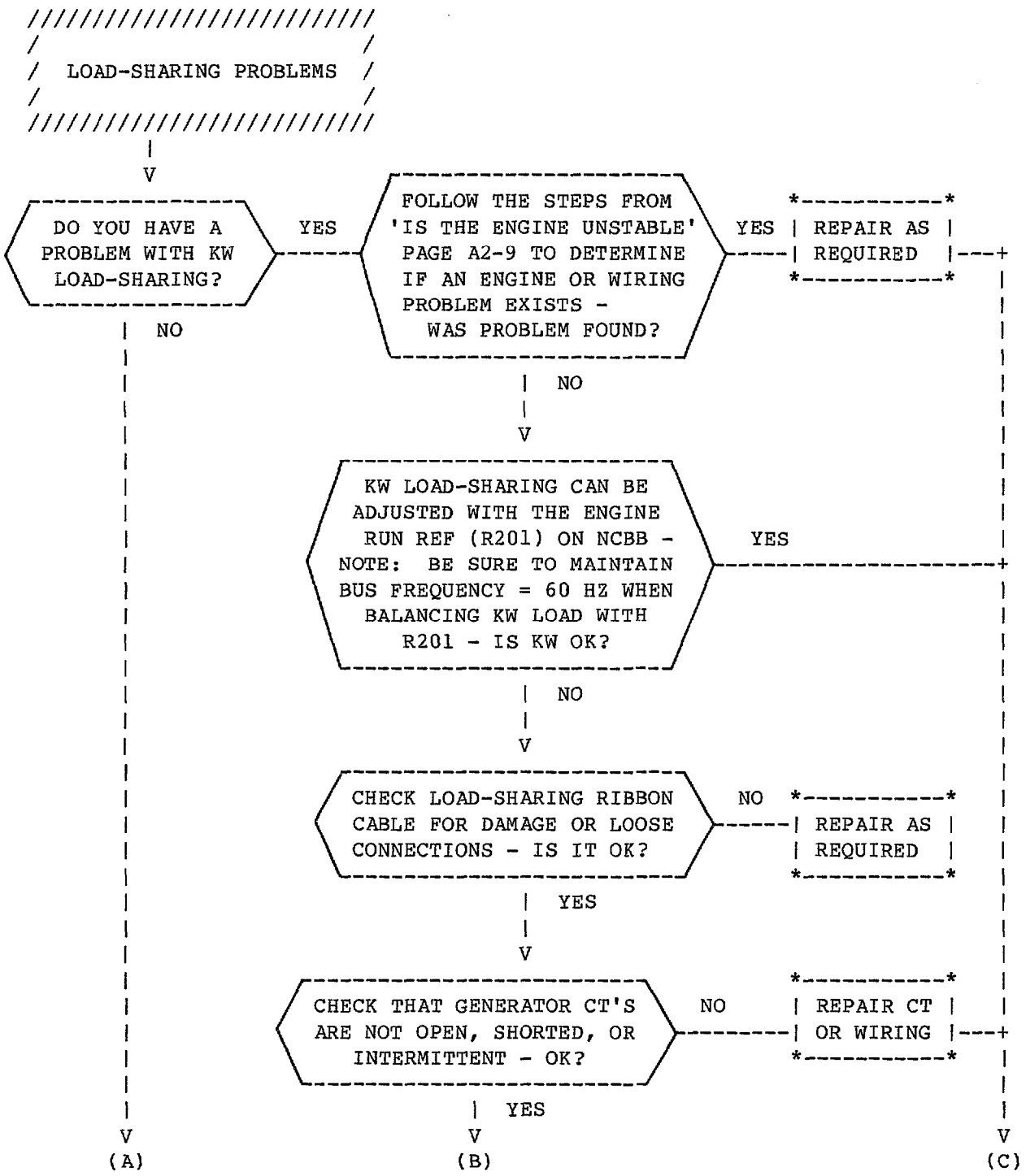
A-2.6 Generator/Engine On-Line Problems



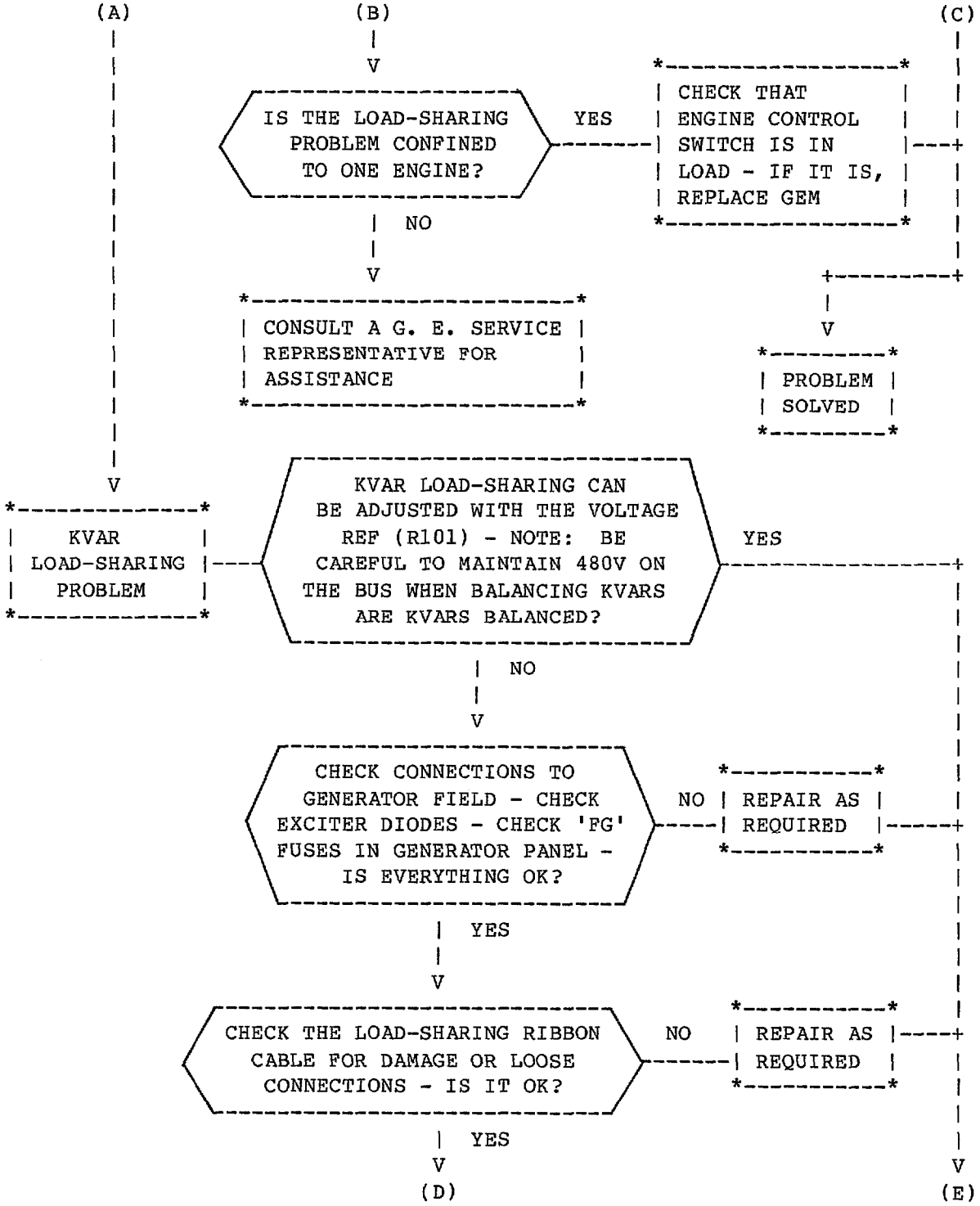
A-2.6 Generator/Engine On-Line Problems (Continued)



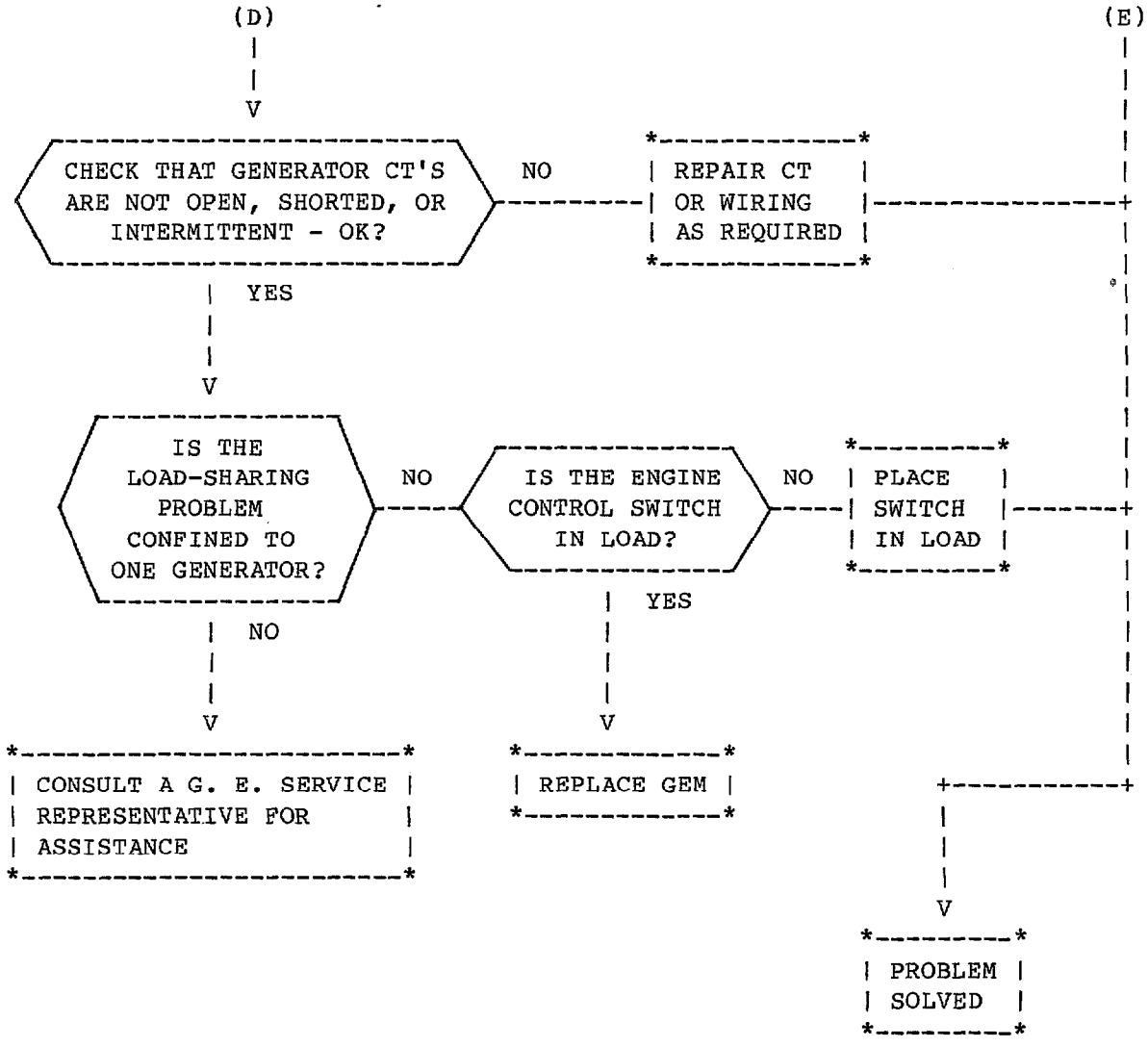
A-2.7 Load-Sharing Problems



A-2.7 Load-Sharing Problems (Continued)

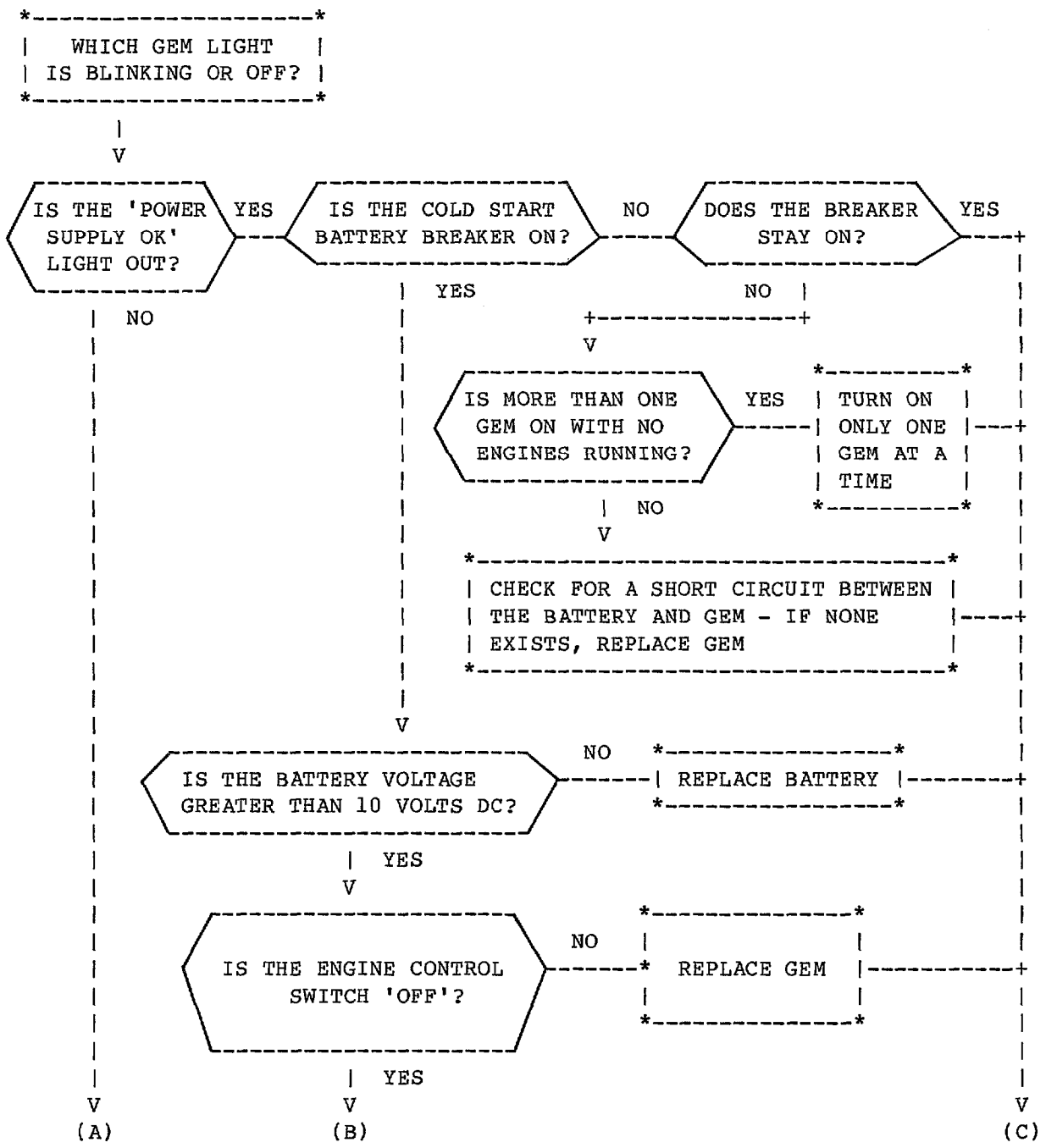


A-2.7 Load-Sharing Problems (Continued)

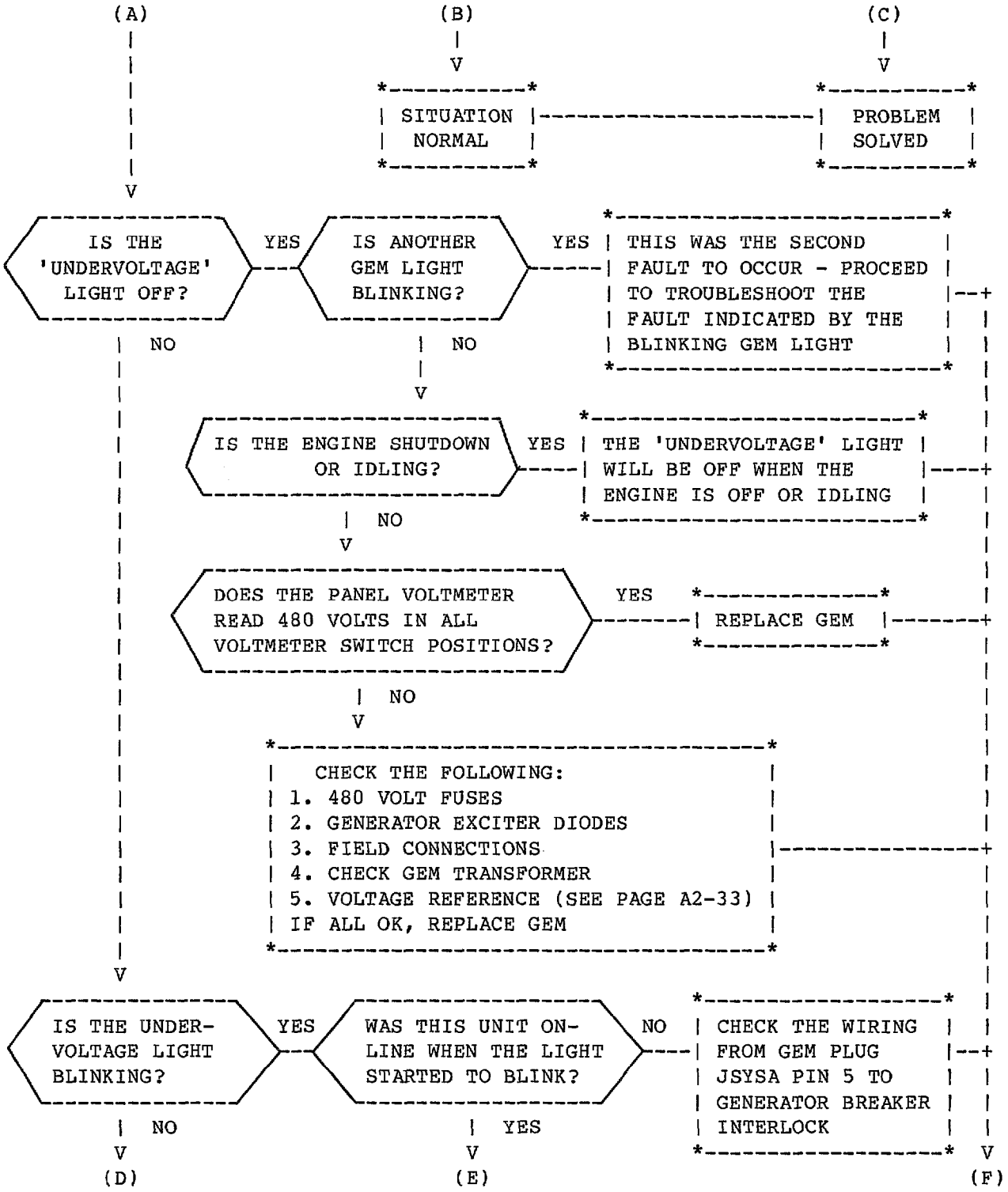


A-2.8 Diagnostic Troubleshooting

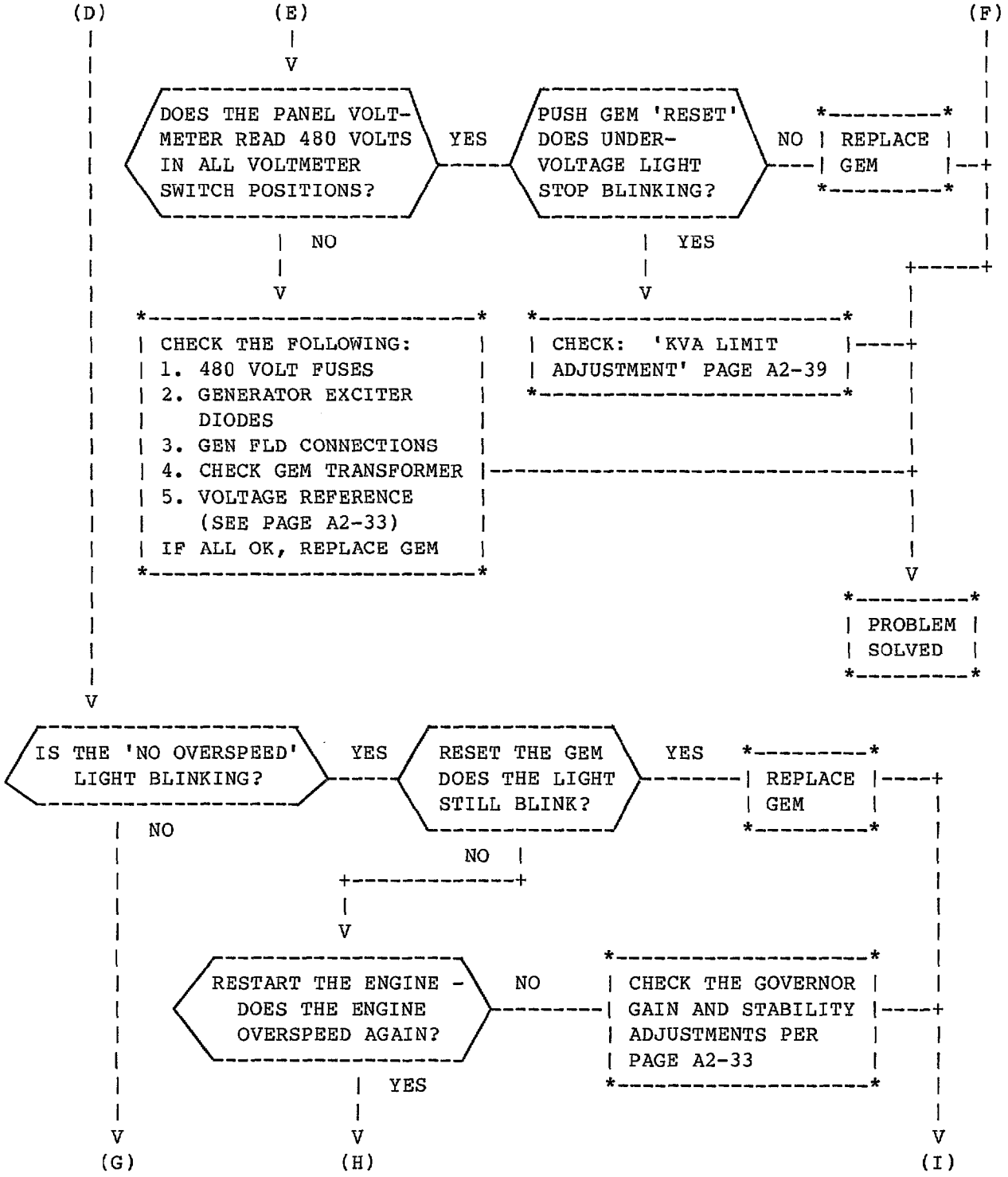
DIAGNOSTIC TROUBLESHOOTING



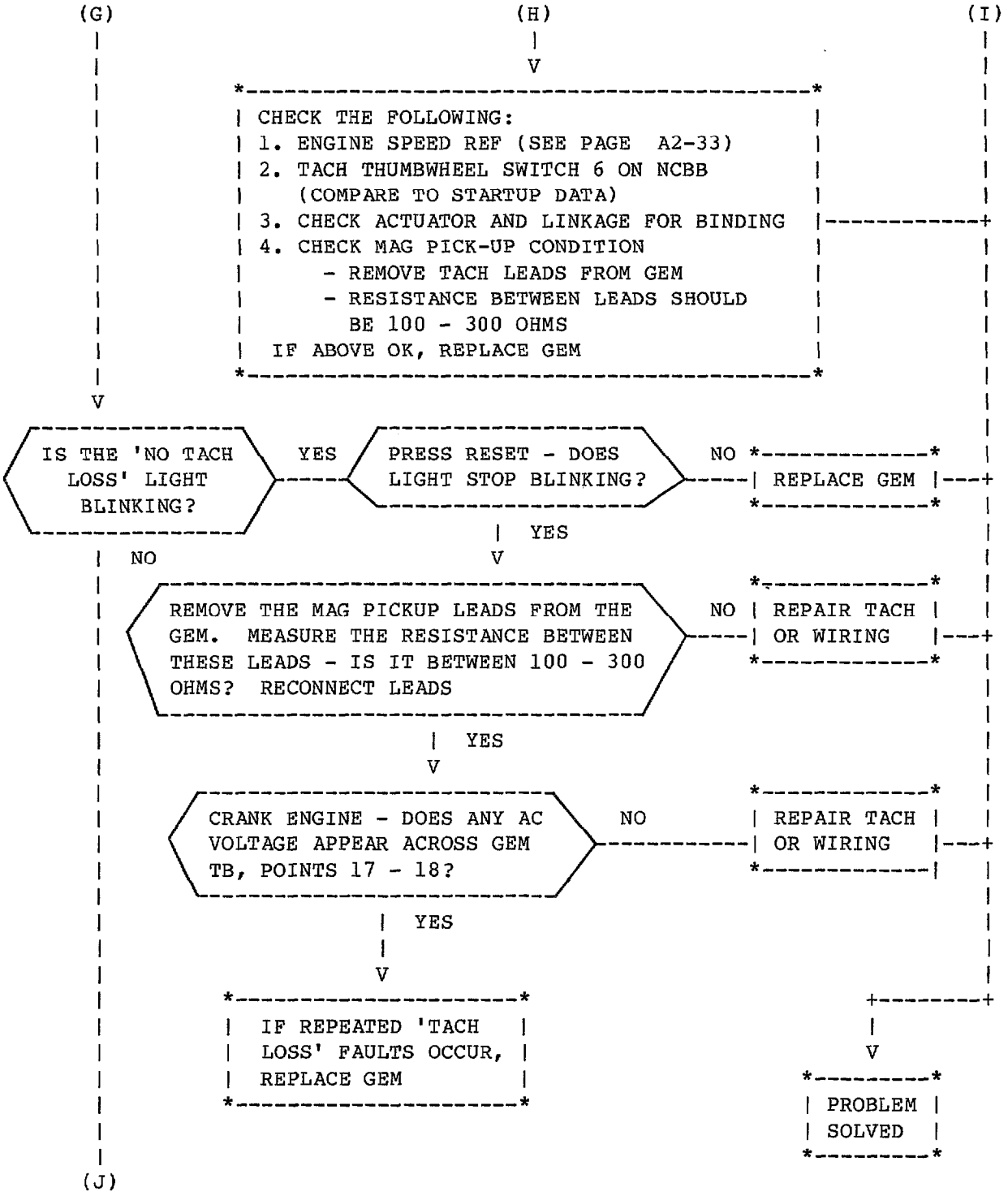
A-2.8 Diagnostic Troubleshooting (Continued)



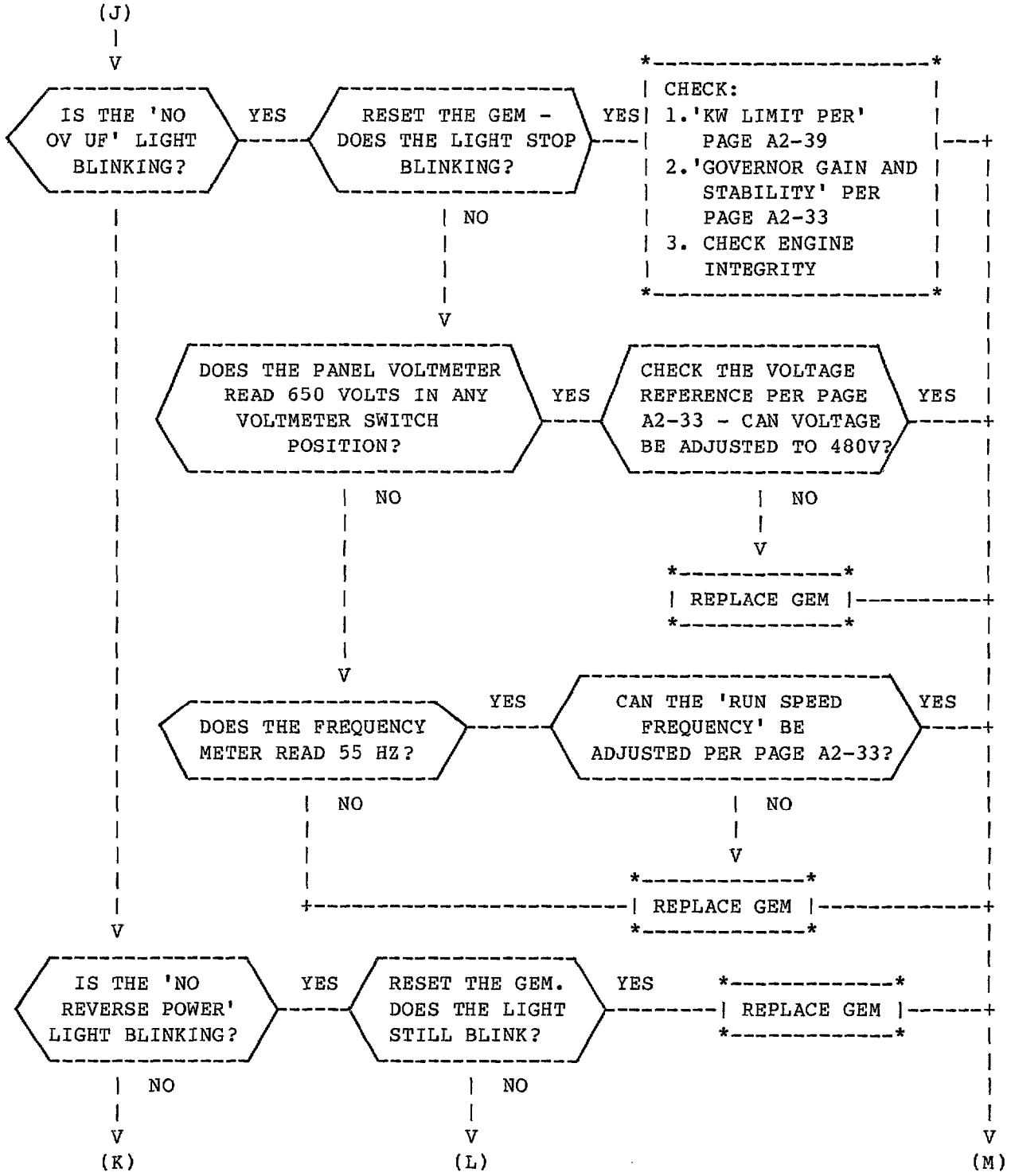
A-2.8 Diagnostic Troubleshooting (Continued)



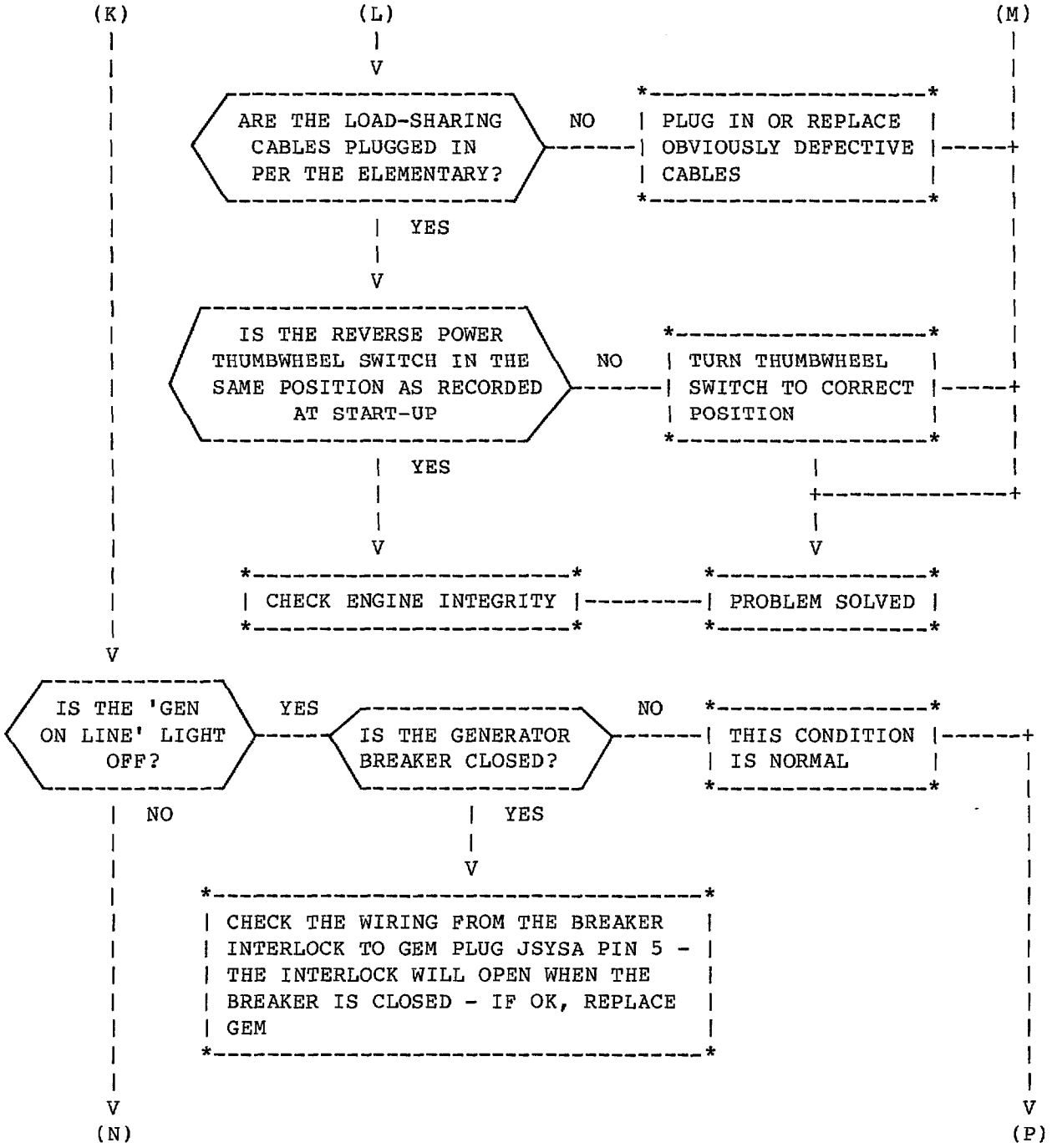
A-2.8 Diagnostic Troubleshooting (Continued)



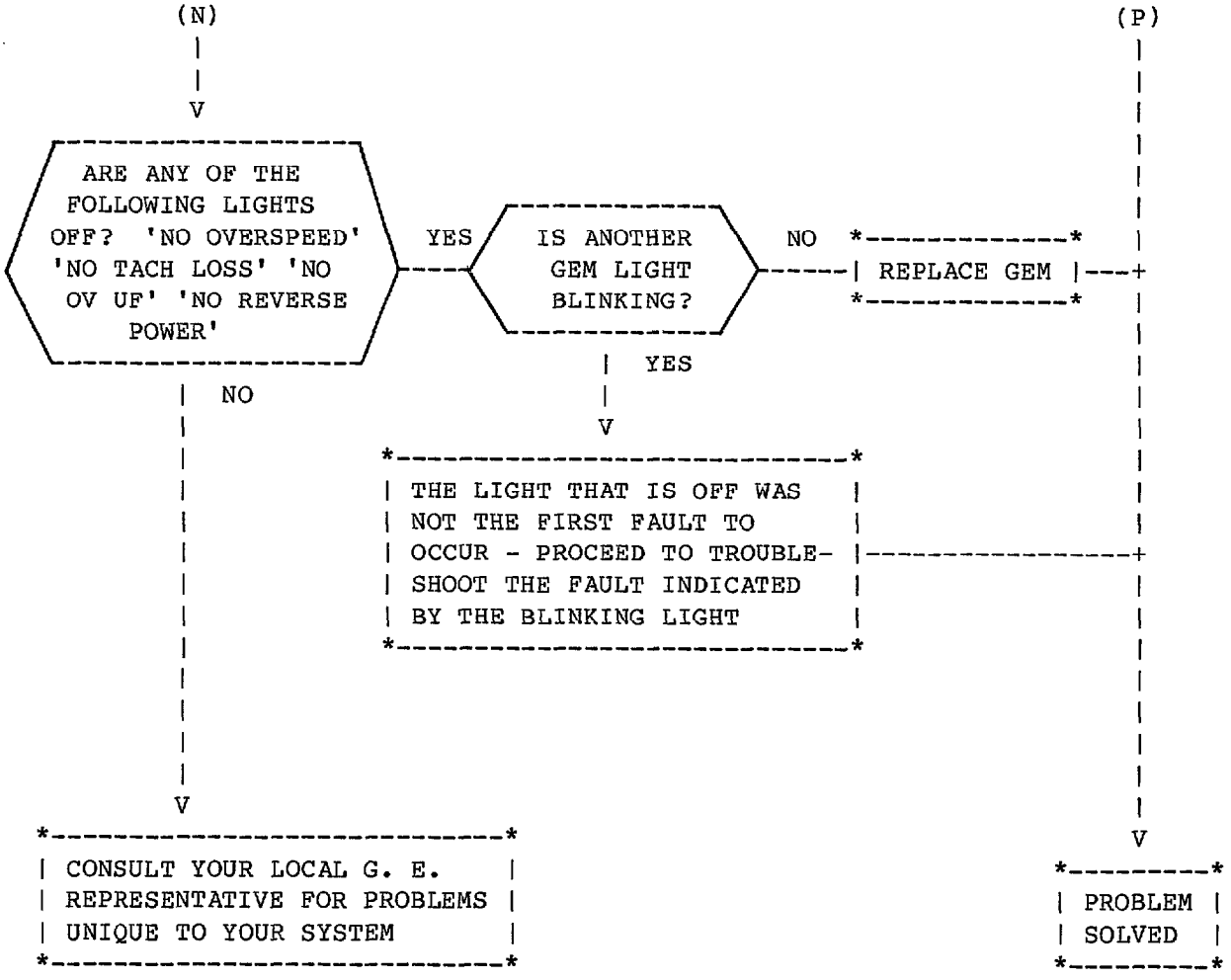
A-2.8 Diagnostic Troubleshooting (Continued)



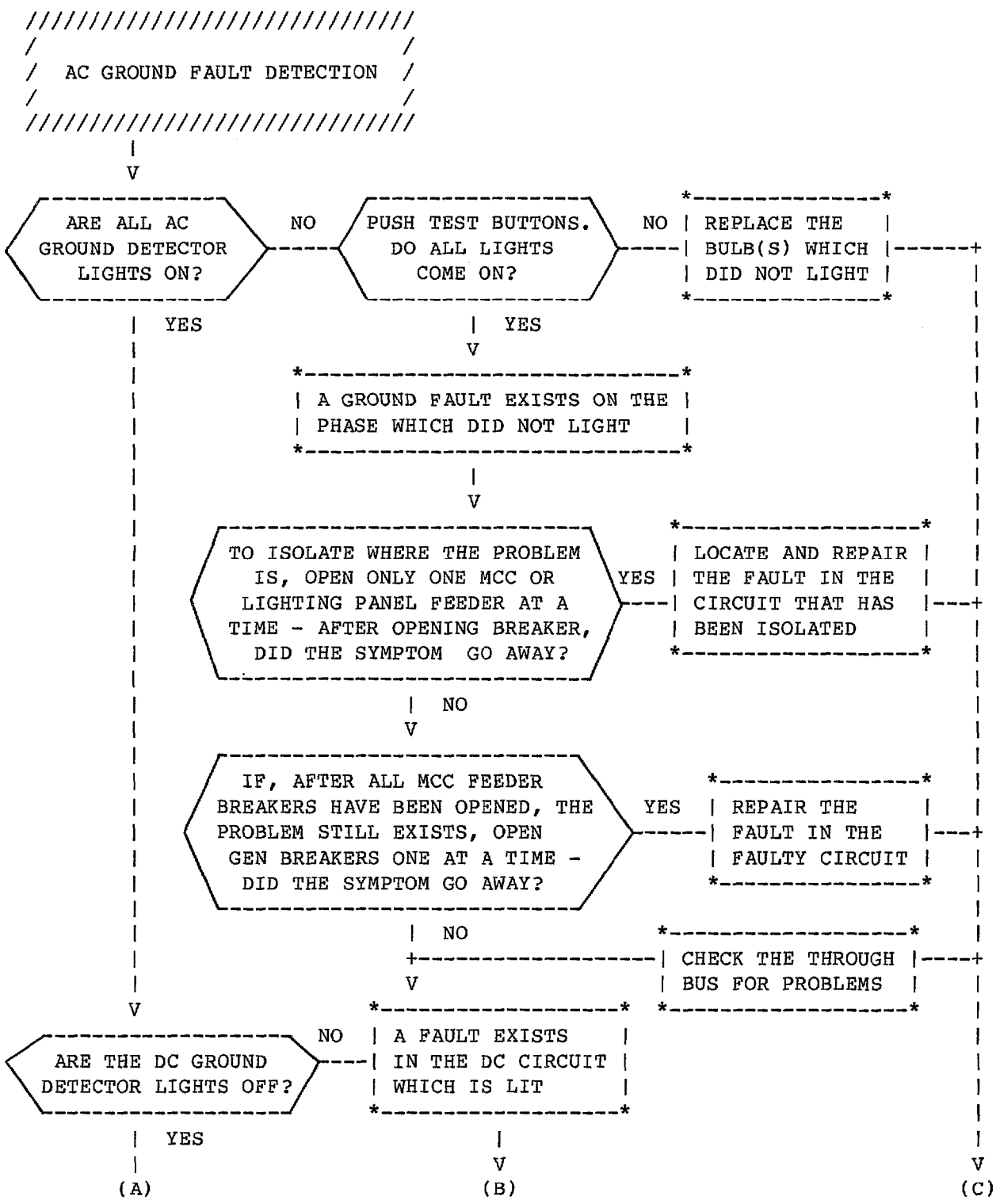
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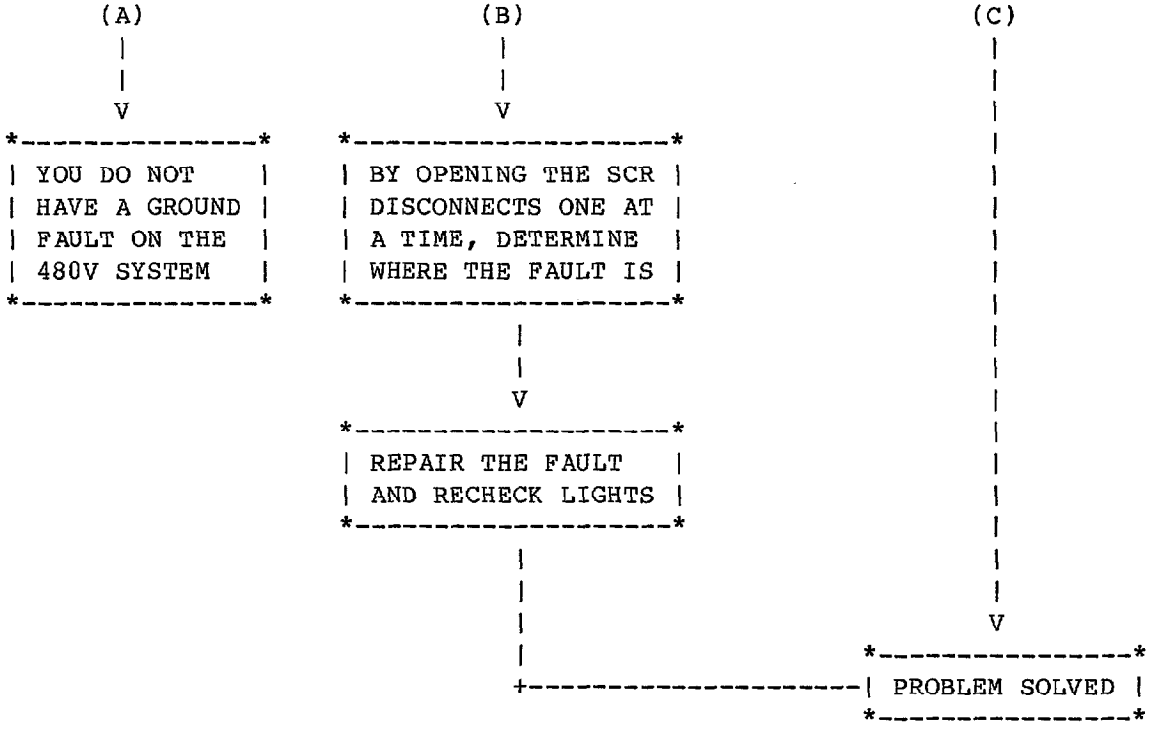
A-2.8 Diagnostic Troubleshooting (Continued)



A-2.9 AC Ground Fault Detection



A-2.9 AC Ground Fault Detection (Continued)



A-2.10 GEM Adjustments

A. ENGINE ADJUSTMENTS

ALL GEM ADJUSTMENTS ARE LOCATED ON THE FRONT DOOR (DS3800NCBB) OF THE GEM. REFER TO FIGURE A-1, PAGE A2-38 TO HELP LOCATE A SPECIFIC ADJUSTMENT.

NOTE: CASUAL TUNE-UP OF THE GEM IS NOT REQUIRED. THIS SECTION IS INCLUDED ONLY TO HELP YOU SOLVE GEM PROBLEMS.

PLEASE KEEP IN MIND THAT REPLACEMENT MODULES CAN BE ADJUSTED BY SIMPLY COMPARING POTS AND SWITCHES TO THE ORIGINAL MODULE OR THE START-UP DATA.

1. IDLE SPEED - TURN ENGINE CONTROL SWITCH TO IDLE. ADJUST 'IDLE REF' POT R102 FOR DESIRED SPEED.

2. RUN SPEED (FREQUENCY) - TURN ENGINE CONTROL SWITCH TO 'RUN.' GENERATOR BREAKER MUST BE OPEN. TURN FREQUENCY 'TRIM POT' TO MIDDLE POSITION. TURN 'FREQUENCY SWITCH' TO DESIRED POSITION. ADJUST POT R105 SO THAT FREQUENCY METER READS 60 HZ.

3. GOVERNOR STABILITY - ADJUST POT R103 TO POSITION 10. NO FURTHER ADJUSTMENT IS REQUIRED.

4. GOVERNOR GAIN - TURN ENGINE CONTROL SWITCH TO 'RUN.' GENERATOR BREAKER MUST BE OPEN. TURN POT R104 CLOCKWISE UNTIL ENGINE BEGINS TO GO UNSTABLE. NEXT, SLOWLY TURN POT R104 COUNTERCLOCKWISE JUST FAR ENOUGH SO THAT ENGINE IS STABLE. NOTE: ENGINE STABILITY CAN BE MONITORED BY WATCHING THE FREQUENCY METER. NEEDLE MOVEMENT SHOULD BE LESS THAN 0.1 HZ.

5. TACH RANGE SWITCH #6 - SET SWITCH TO SAME POSITION AS RECORDED AT START-UP. NOTE: TURN ENGINE CONTROL SWITCH TO 'OFF' WHEN MAKING THIS ADJUSTMENT.

6. REVERSE POWER SWITCH #7 - THE REVERSE POWER SWITCH SETS THE LEVEL AND TIME DELAY REQUIRED TO TRIP THE GENERATOR BREAKER ON A REVERSE POWER FAULT. THE TYPICAL SETTING FOR THIS SWITCH IS POSITION 'A', WHERE 10% OF REVERSE POWER WILL CAUSE GENERATOR BREAKER TO TRIP IN APPROXIMATELY 3.5 SEC.

NOTE 1: THE BREAKER WILL TRIP FASTER FOR THE SAME LEVEL OF REVERSE POWER AS THE THUMBWHEEL SWITCH IS MOVED TOWARD POSITION '0'. IT WILL TAKE LONGER TO TRIP AS THE THUMBWHEEL SWITCH IS MOVED TOWARDS POSITION 'F'.

NOTE 2: TURN ENGINE CONTROL SWITCH TO 'OFF' WHEN MAKING THIS ADJUSTMENT.

A-2.10 GEM Adjustments (Continued)

A. ENGINE ADJUSTMENTS (Continued)

7. KW LOAD-SHARING - TURN THE ENGINE CONTROL SWITCH ON ALL ON-LINE UNITS TO 'LOAD'. ADJUST POT R105 TO BALANCE THE KW LOAD. BE CAREFUL TO MAINTAIN 60 HZ ON THE FREQUENCY METER WHEN MAKING THIS ADJUSTMENT.

B. GENERATOR ADJUSTMENTS

ALL GEM ADJUSTMENTS ARE LOCATED ON THE FRONT DOOR (DS3800NCBB) OF THE GEM. REFER TO FIGURE A-1, PAGE A2-38 TO HELP LOCATE A SPECIFIC ADJUSTMENT.

NOTE: CASUAL TUNE-UP OF THE GEM IS NOT REQUIRED. THIS SECTION IS INCLUDED ONLY TO HELP YOU SOLVE GEM PROBLEMS.

PLEASE KEEP IN MIND THAT REPLACEMENT MODULES CAN BE ADJUSTED BY SIMPLY COMPARING POTS AND SWITCHES TO THE ORIGINAL MODULE OR THE START-UP DATA.

1. VOLTAGE REFERENCE - ENGINE MUST BE AT RATED SPEED. GENERATOR BREAKER MUST BE OPEN. ADJUST POT R101 FOR EXACTLY 480 VOLTS.

2. VOLTAGE GAIN - ADJUST POT R114 TO THE VALUE RECORDED AT START-UP. TYPICAL SETTING = 40.

3. VOLTAGE STABILITY - ADJUST POT R100 TO THE VALUE RECORDED AT START-UP.
TYPICAL SETTING = 80.

4. GENERATOR FIELD HEATING:

(a) ENGINE MUST BE AT IDLE SPEED.
(b) USE A DC VOLTMETER (ON THE 1 VOLT SCALE) TO MEASURE THE VOLTAGE DIRECTLY ACROSS R1 ON XPES.
(c) DETERMINE THE GENERATOR FULL LOAD FIELD EXCITER CURRENT FROM THE GENERATOR NAMEPLATE.
(d) CALCULATE:

$$VTP10 = \frac{52 \times \text{VOLTAGE ACROSS R1 ON XPES}}{\text{GENERATOR FULL LOAD FIELD CURRENT}}$$

(e) OPEN THE GEM DOOR AND LOCATE TP10 ON THE HFPA CARD. MEASURE THE VOLTAGE AT TP10 WITH A DC VOLTMETER (10 VOLT SCALE).

NOTE: THE COMMON PROBE OF THE VOLTMETER SHOULD BE ATTACHED TO CTB-2.

A-2.10 GEM Adjustments (Continued)

B. GENERATOR ADJUSTMENTS (Continued)

(f) ADJUST THE FIELD HEATING GAIN (POT R113 ON NCBB) SO THAT THE VOLTAGE ON HFPA TP10 IS EQUIVALENT TO THE VALUE CALCULATED IN STEP 4.

5. KVAR LOAD-SHARING - TURN THE ENGINE CONTROL SWITCH ON ALL ON-LINE UNITS TO 'LOAD'. ADJUST POT R101 TO BALANCE THE KVAR LOAD. BE CAREFUL TO MAINTAIN 480 VOLTS ON THE BUS WHEN MAKING THIS ADJUSTMENT.

C. METERING ADJUSTMENTS

ALL GEM ADJUSTMENTS ARE LOCATED ON THE FRONT DOOR (DS3800NCBB) OF THE GEM. REFER TO FIGURE A-1, PAGE A2-38 TO HELP LOCATE A SPECIFIC ADJUSTMENT.

NOTE: CASUAL TUNE-UP OF THE GEM IS NOT REQUIRED. THIS SECTION IS INCLUDED ONLY TO HELP YOU SOLVE GEM PROBLEMS.

PLEASE KEEP IN MIND THAT REPLACEMENT MODULES CAN BE ADJUSTED BY SIMPLY COMPARING POTS AND SWITCHES TO THE ORIGINAL MODULE OR THE START-UP DATA.

1. PHASE 1 AMMETER GAIN:

(a) PUT ONE GENERATOR ON-LINE AND TURN THE ENGINE CONTROL SWITCH TO 'LOAD.'

(b) WITH AN AC VOLTMETER (1 VOLT SCALE), MEASURE THE VOLTAGE DIRECTLY ACROSS R3 ON XPES.

(c) DETERMINE THE CT PRIMARY RATING FROM SHEET 0F IN THE SYSTEM ELEMENTARY.

(d) CALCULATE:

GEN PHASE 1 AMPS = (VOLTAGE ACROSS R3) X (CT PRIMARY RATING)

(e) PLACE AMMETER SWITCH IN 'PHASE 1'. ADJUST R106 SO THAT THE GENERATOR AMMETER READS THE VALUE CALCULATED IN STEP C.1.(d).

A-2.10 GEM Adjustments (Continued)

1. PHASE 1 AMMETER GAIN: (Continued)

   (f) EXAMPLE: PUT ONE UNIT ON-LINE

   VOLTAGE ACROSS R3 = .2 VAC
   CT RATIO = 1600/5
   GEN PHASE 1 AMPS = .2 X 1600 = 320

   TURN AMMETER SW TO PHASE 1.
   ADJUST R106 SO THAT AMMETER READS 320 AMPS.

2. PHASE 3 AMMETER GAIN:

   (a) PUT ONE GENERATOR ON-LINE AND TURN THE ENGINE CONTROL SWITCH TO 'LOAD.'

   (b) WITH AN AC VOLTMETER (1 VOLT SCALE) MEASURE THE VOLTAGE DIRECTLY ACROSS R4 ON XPES.

   (c) DETERMINE THE CT PRIMARY RATING.

   (d) CALCULATE:

   GEN PHASE 3 AMPS = (VOLTAGE ACROSS R4) X (CT PRIMARY RATING)

   (e) PLACE AMMETER SWITCH IN 'PHASE 3'. ADJUST R107 SO THAT THE GENERATOR AMMETER READS THE VALUE CALCULATED IN STEP 4.

3. KW GAIN:

   (a) PUT ONE GENERATOR ON LINE. TURN THE ENGINE CONTROL SWITCH TO LOAD.

   (b) RECORD GENERATOR AMPS.

   (c) RECORD GENERATOR VOLTS.

   (d) CALCULATE:

$$\text{GEN KVA} = \frac{\sqrt{3} \times (\text{GEN VOLTS}) \times (\text{GEN AMPS})}{1000}$$

   (e) RECORD GENERATOR KVARS.

   (f) CALCULATE:

$$\sqrt{(\text{GEN KVA})^2 - (\text{GEN KVARS})^2}$$

A-2.10 GEM Adjustments (Continued)

3. KW GAIN (Continued):

   (g) ADJUST POT R205 ON NCBB SO THAT KW METER READS THE VALUE CALCULATED IN STEP 6.

4. FREQUENCY GAIN - ADJUSTMENT OF THIS POT IS NOT REQUIRED.

A-2.10 GEM Adjustments (Continued)

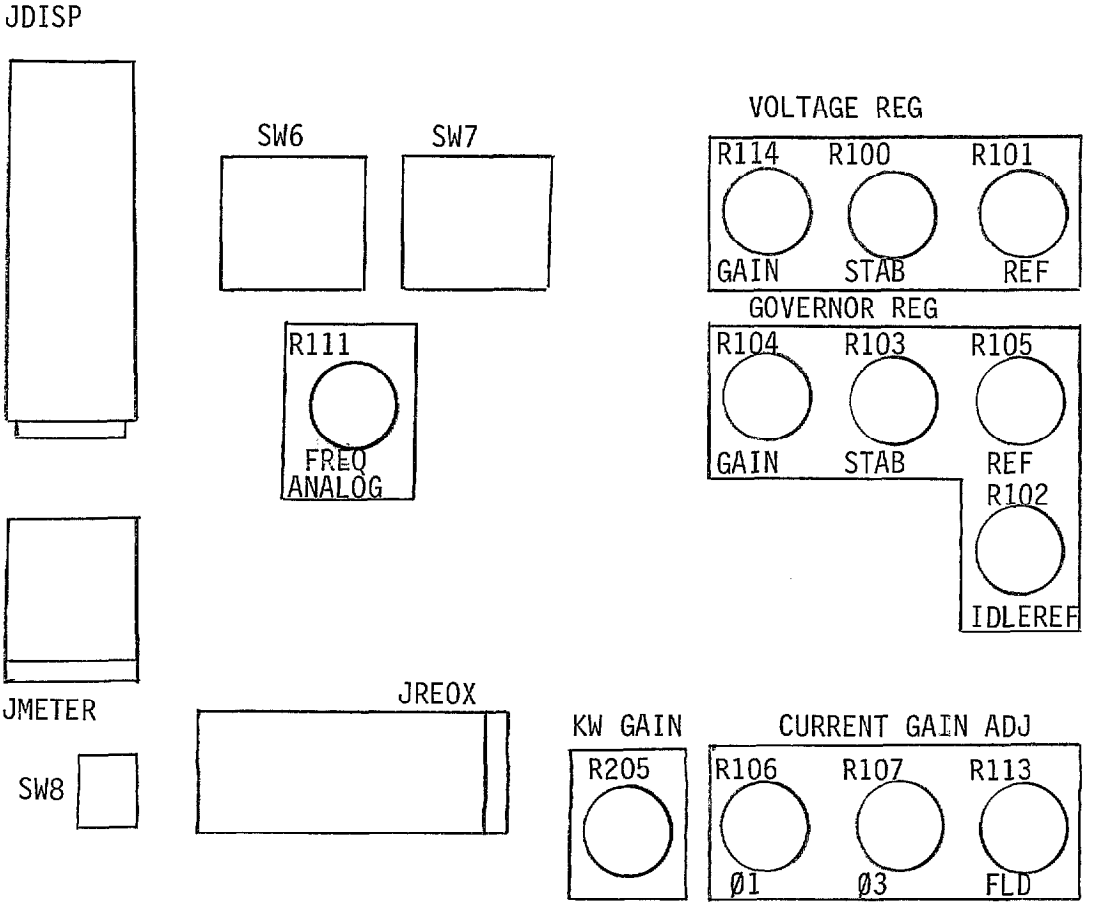


FIGURE A-1: - GEM ADJUSTMENT LOCATIONS (DS3800NCBB)

A-2.11 Power Limit Adjustments

NOTE: TO MAKE THE ADJUSTMENTS IN THIS SECTION, YOU WILL NEED A TEST LEAD WITH A 'BANANA PLUG' ON ONE END AND AN 'E-Z CLIP' ON THE OTHER.

A. KW LIMIT

1. LOCATE THE KVA MODULE.

2. CLOSE ANY PCP BREAKER.

3. REMOVE PLUG 'JCX' FROM A GENERATOR MODULE WHICH IS NOT IN USE AND IS TURNED OFF. LET'S ASSUME THIS IS GEM #1.

4. CONNECT YOUR TEST LEAD FROM THE 'TEST' JACK ON THE AC INTERFACE MODULE TO TEST POINT 9 ON THE HFPA CARD IN GEM 1.

5. ADJUST POT 4 (ON THE CARD IN SLOT 0C IN THE KW MODULE) SO THAT THE KW METER ON GENERATOR PANEL #4 READS THE RATED KW OF ONE ENGINE.

6. ADJUST THE KW BIAS (POT 1, CARD SLOT 0C, KW MODULE) SO THAT THE 'LIMIT' TEST JACK ON THE KVA MODULE MODULE EQUALS -5.0 VDC.

7. REMOVE JUMPER FROM HFPA 9 ON GEM 1 AND REPLACE JCX PLUG.

B. KVA LIMIT

1. TURN ON ANY PCP.

2. CALCULATE THE KVA LIMIT SIGNAL EXPECTED FROM THE GEM MODULE WITH THE GEM AT RATED LOAD AS FOLLOWS:

$$V_{KVA} = \frac{-1 \text{ VOLT}}{200 \text{ AMPS}} \times \text{RATED GENERATOR AMPS}$$

3. ADJUST THE VOLTAGE AT THE 'TEST' JACK ON THE AC INTERFACE MODULE TO THE VALUE CALCULATED (V_{KVA}) IN STEP 2. THIS VALUE MUST BE NEGATIVE AND CAN BE OBTAINED BY ADJUSTING POT 4, CARD SLOT 0C, IN THE KW MODULE.

4. REMOVE THE WIRE CONNECTED TO THE JDC-4 'STAB ON' ON ANY GEM MODULE.

5. JUMPER THE 'TEST' JACK VOLTAGE (STEP 3) TO THE WIRE REMOVED IN STEP 4.

6. ADJUST THE KVA BIAS (POT 3, CARD SLOT 0C, KW MODULE SO THAT THE 'LIMIT' TEST JACK EQUALS -5.0 VDC.

7. REMOVE JUMPER FROM THE 'TEST' JACK AND RECONNECT THE ORIGINAL WIRE BACK TO THE JDC-4 STAB-ON.

NOTES